Evaluating multi-season occupancy models with autocorrelation fitted to heterogeneous datasets

A Supporting Information A - Simulated data designs

Table A.1: Spatial and temporal autocorrelation scenarios of Doser & Stoudt (2024). We refer them to 'sub scenarios' as their scenarios were nested within each one of our studies and data design scenarios.

Sub scenario	σ^2	ϕ	ρ	σ^2_T
1	0.3	15.00	0.5	0.3
2	1.5	15.00	0.5	0.3
3	0.3	3.75	0.5	0.3
4	1.5	3.75	0.5	0.3
5	0.3	15.00	0.9	0.3
6	1.5	15.00	0.9	0.3
7	0.3	3.75	0.9	0.3
8	1.5	3.75	0.9	0.3
9	0.3	15.00	0.5	1.5
10	1.5	15.00	0.5	1.5
11	0.3	3.75	0.5	1.5
12	1.5	3.75	0.5	1.5
13	0.3	15.00	0.9	1.5
14	1.5	15.00	0.9	1.5
15	0.3	3.75	0.9	1.5
16	1.5	3.75	0.9	1.5

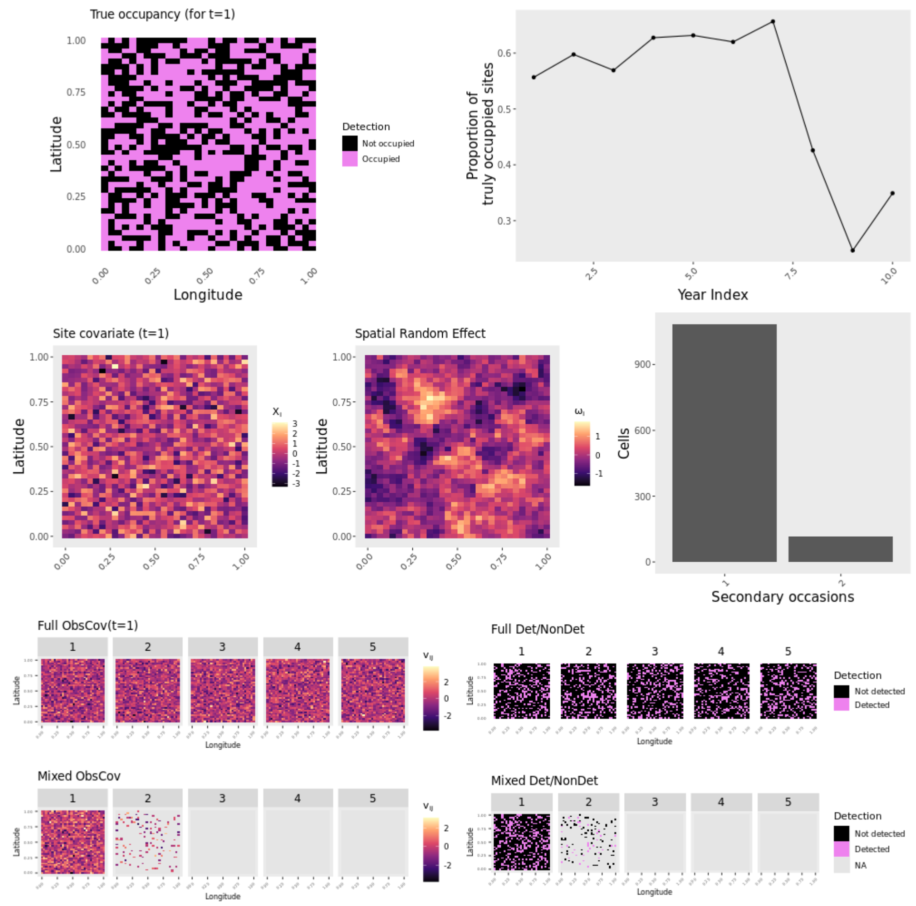

Figure A.1: Design of one simulated dataset in D&S (for one primary occasion $t = 1$) (study-scenario 1-0). Data shown in the upper row: True site occupancy in the first year (i.e. $z_{i,t=1}$), proportion of truly occupied sites over the years ($z_t = \frac{1}{I}\sum_{i=I}^{I} z_{it}$), for η_t estimated with $\rho = 0.5$ and $\sigma_T^2 = 0.3$. Middle row: Spatial structure of the site covariate (left, X_it) and spatial random effect (middle, for ω_i estimated with $\phi = 3.75$ and $\sigma^2 = 0.3$). Two-group distribution of repeats is show in the right, with sampling in $j = 2$ obeying a Bernoulli distribution with $p = 0.1$. Bottom row: Observation covariate (v_{itj}, left) and the detection and non-detection data (y_{itj}, right) for the full data ($J = 5$, top) and $J = 2$ secondary occasions (bottom). The subscripts t were omitted because only data simulated for $t = 1$ are shown.

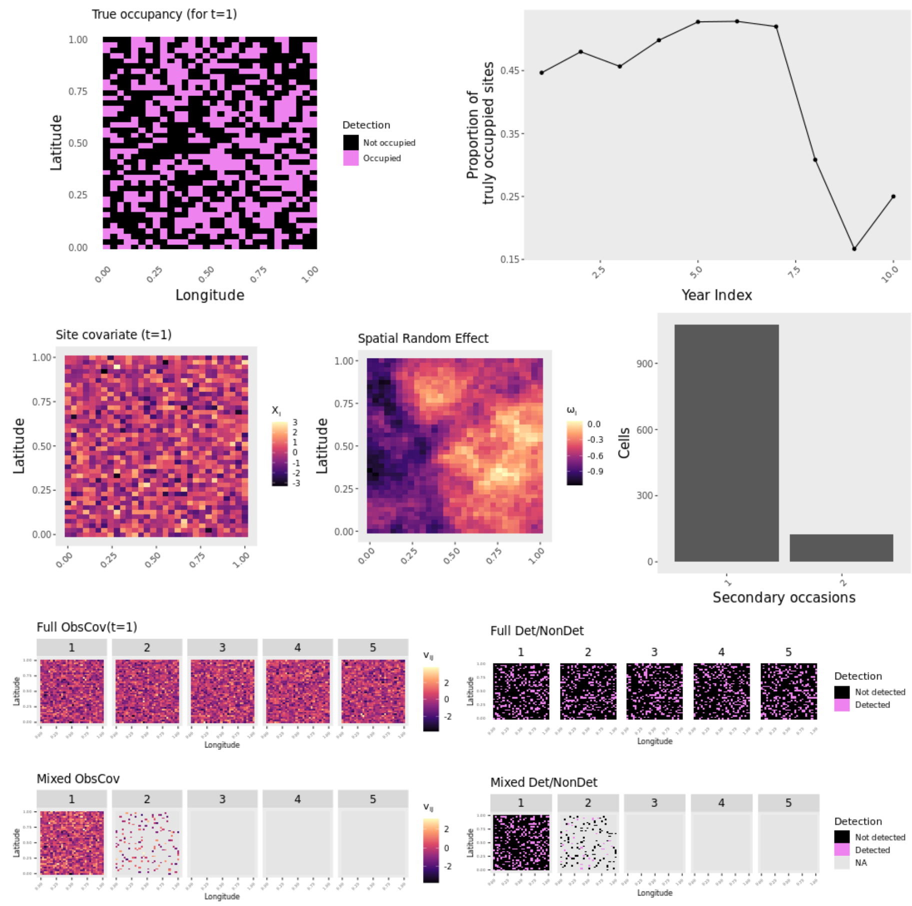

Figure A.2: Design of one simulated dataset in D&S (for one primary occasion $t = 1$) under high spatial autocorrelation (slower spatial decay $\phi = 0.5$) (study-scenario 1-1). Data shown in the upper row: True site occupancy in the first year (i.e. $z_{i,t=1}$), proportion of truly occupied sites over the years ($z_t = \frac{1}{I}\sum_{i=I}^{I} z_{it}$), for η_t estimated with $\rho = 0.5$ and $\sigma_T^2 = 0.3$. Middle row: Spatial structure of the site covariate (X_i, left) and spatial random effect (middle, for ω_i estimated with $\phi = 0.5$ and $\sigma^2 = 0.3$). Two-group distribution of repeats is show in the right, with sampling in $j = 2$ obeying a Bernoulli distribution with $p = 0.1$. Bottom row: Observation covariate (v_{itj}, left) and the detection and non-detection data (y_{itj}, right) for the full data ($J = 5$, top) and $J = 2$ secondary occasions (bottom). The subscripts t were omitted because only data simulated for $t = 1$ are shown.

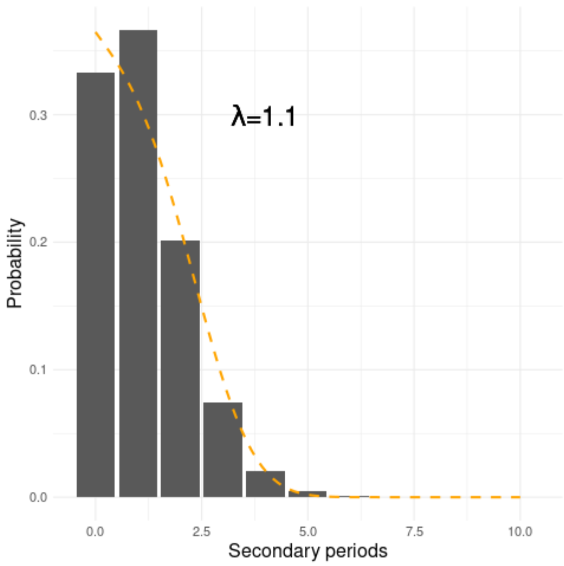

Figure A.3: Expected number of secondary occasions (x-axis) across sites (y-axis) within one primary occasion, as produced by a Poisson distribution with average $\lambda = 1.1$. The values along the y-axis represent the probability mass function of a Poisson distribution. The orange line represents a quadratic spline of a Poisson Generalized Linear Model fitted to the probability distribution data.

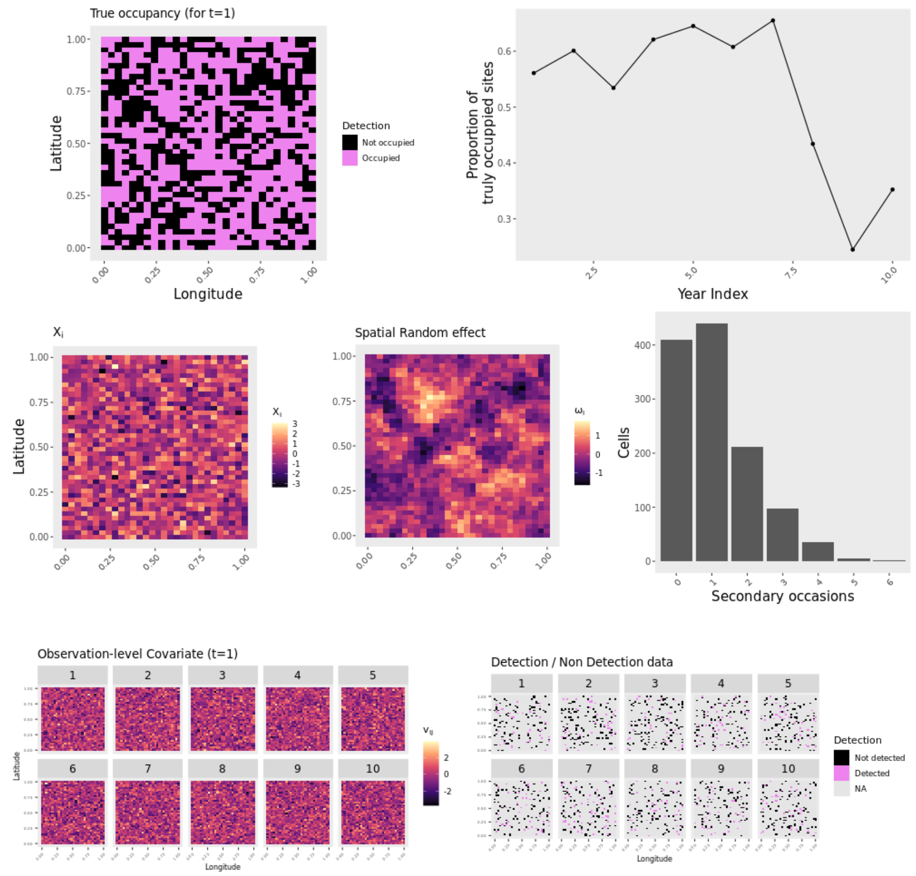

Figure A.4: Data design of our study 2 scenario zero (2-0). The design was based on a Poisson distribution with average $\lambda = 1.1$ to spread J secondary occasions across sites and primary occasions (middle row, right). Bottom row: Observation covariate (v_{itj}, left) and the detection and non-detection data (y_{itj}, right) for $J = 10$ secondary occasions. The subscripts t were omitted because only data simulated for $t = 1$ are shown.

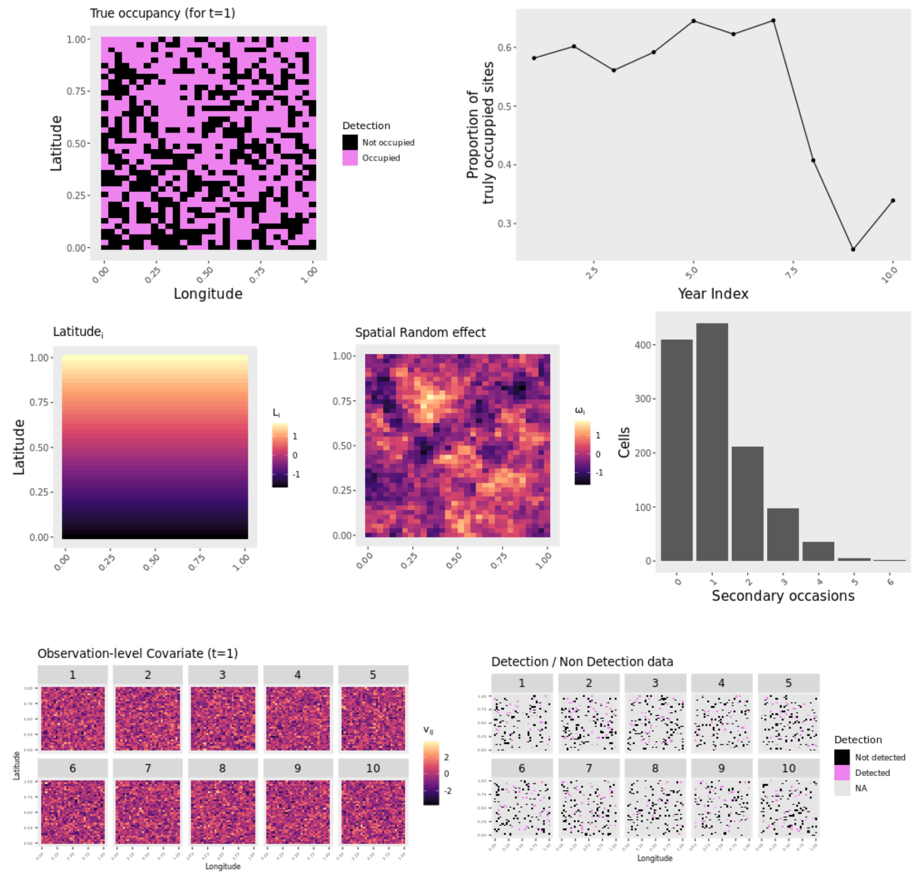

Figure A.5: Data design of our study 2 scenario 1 (2-1). The design was based on a Poisson distribution with average $\lambda = 1.1$ to spread J secondary occasions across sites and primary occasions (middle row, right), and latitude L_i as the single occupancy predictor. The subscripts t were omitted because only data simulated for $t = 1$ are shown.

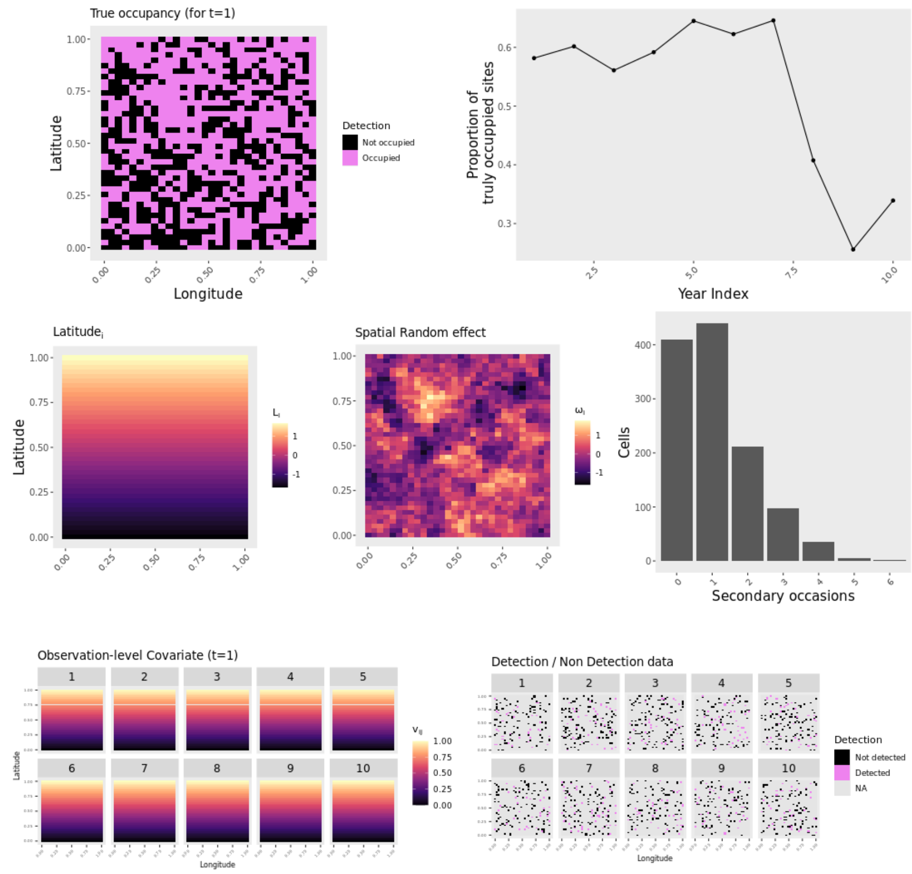

Figure A.6: Data design of our study 2 scenario 2 (2-2). The design was based on a Poisson distribution with average $\lambda = 1.1$ to spread J secondary occasions across sites and primary occasions (middle row, right), and latitude L_i as occupancy and detection predictor. The subscripts t were omitted because only data simulated for $t = 1$ are shown.

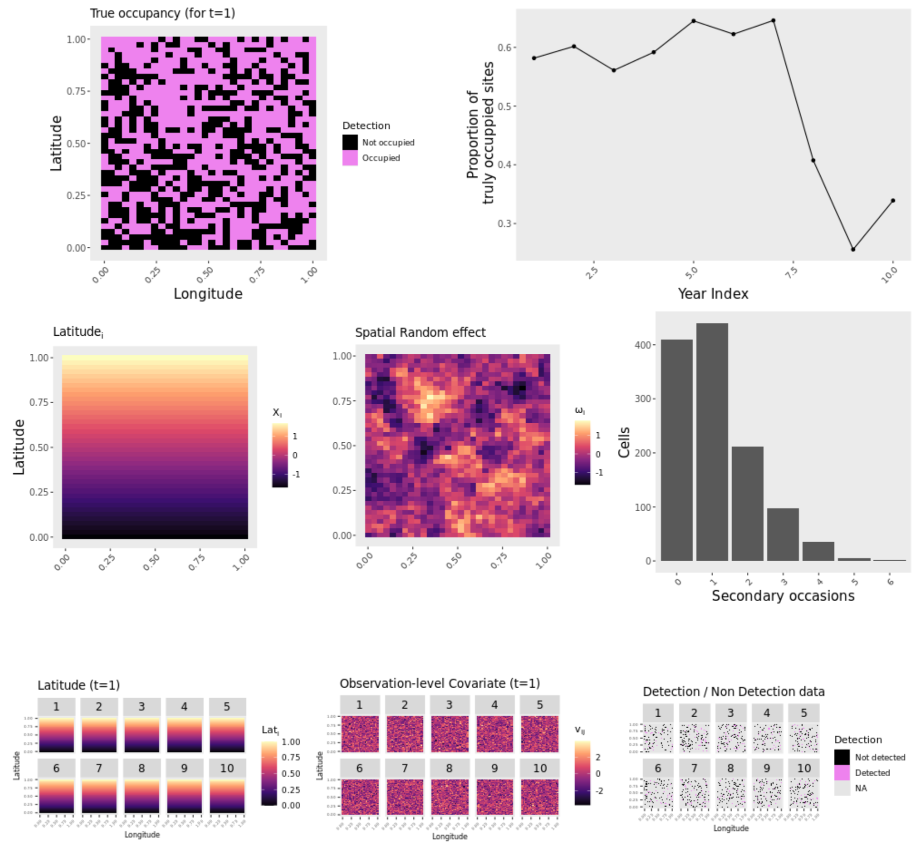

Figure A.7: Data design of our study 2 scenario 3 (2-3). The design was based on a Poisson distribution with average $\lambda = 1.1$ to spread J secondary occasions across sites and primary occasions (middle row, right), and latitude L_i and v_{itj} as detection predictors. The subscripts t were omitted because only data simulated for $t = 1$ are shown.

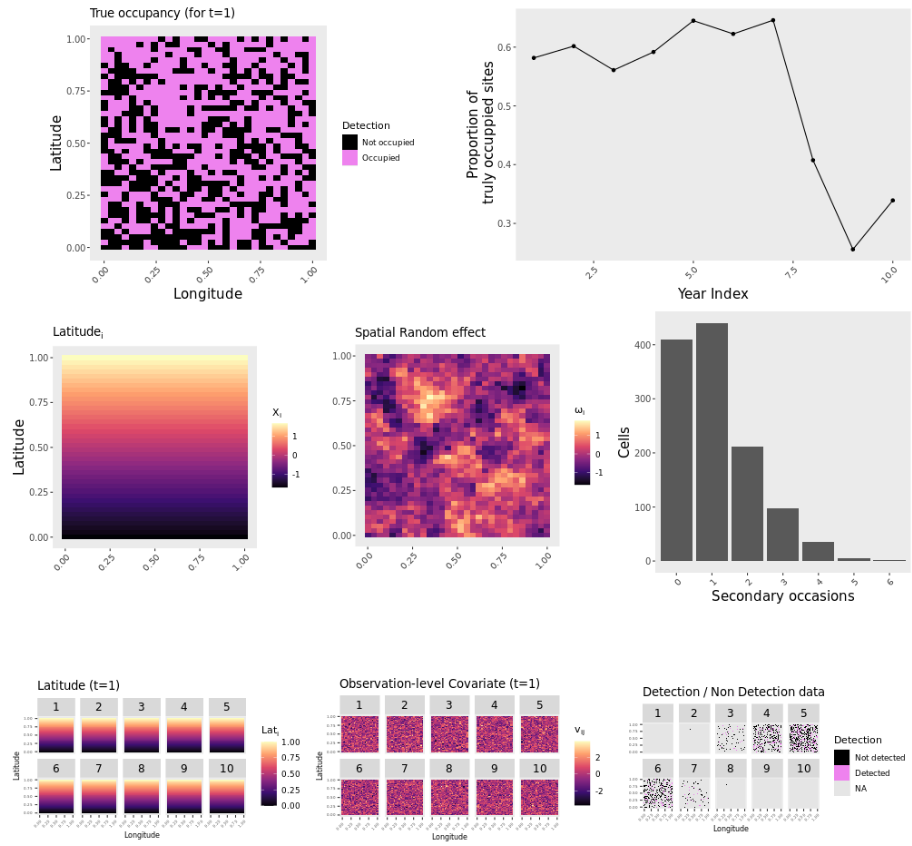

Figure A.8: Data design of our study 3 scenario 1 (3-1). The design was based on a Poisson distribution with average $\lambda = 1.1$ to spread J secondary occasions across sites and primary occasions (middle row, right), latitude L_i and v_{itj} as detection predictors, and phenology effect on the observation process (bottom row). The subscripts t were omitted because only data simulated for $t = 1$ are shown.

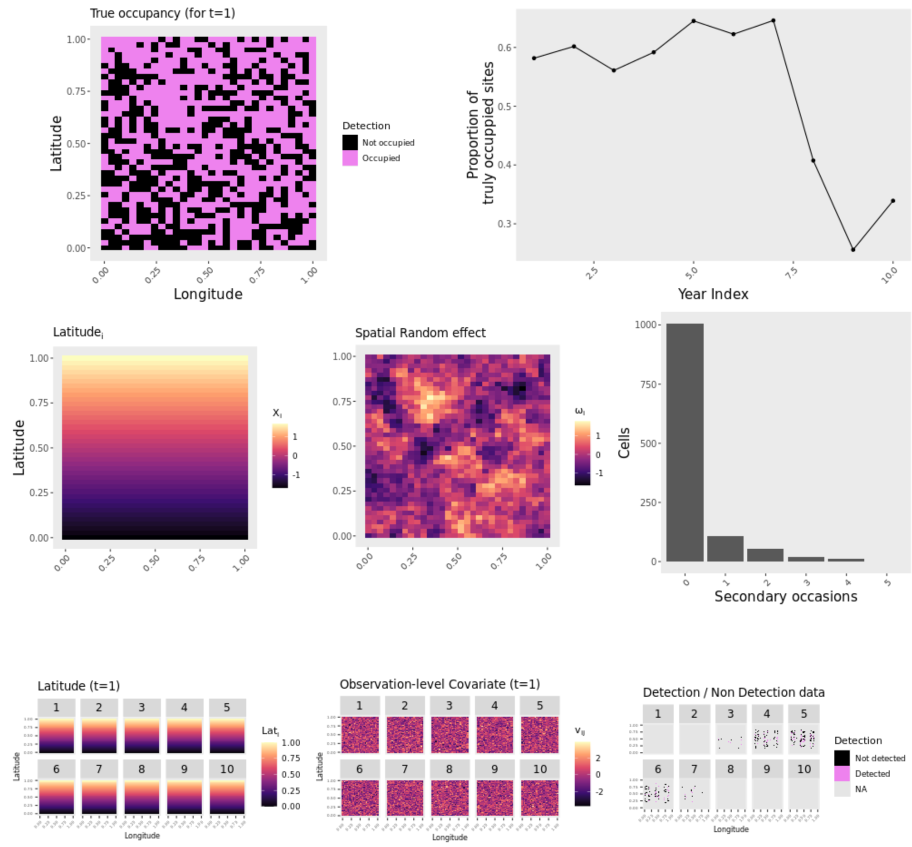

Figure A.9: Data design of our study 3 scenario 2 (3-2). The design was based on a Poisson distribution with average $\lambda = 1.1$ to spread J secondary occasions across sites and primary occasions (middle row, right), latitude L_i and v_{itj} as detection predictors, phenology and spatial spot grouping observations in space and time (bottom row). Note that spatial gaps are generated in this scenario. The subscripts t were omitted because only data simulated for $t = 1$ are shown.

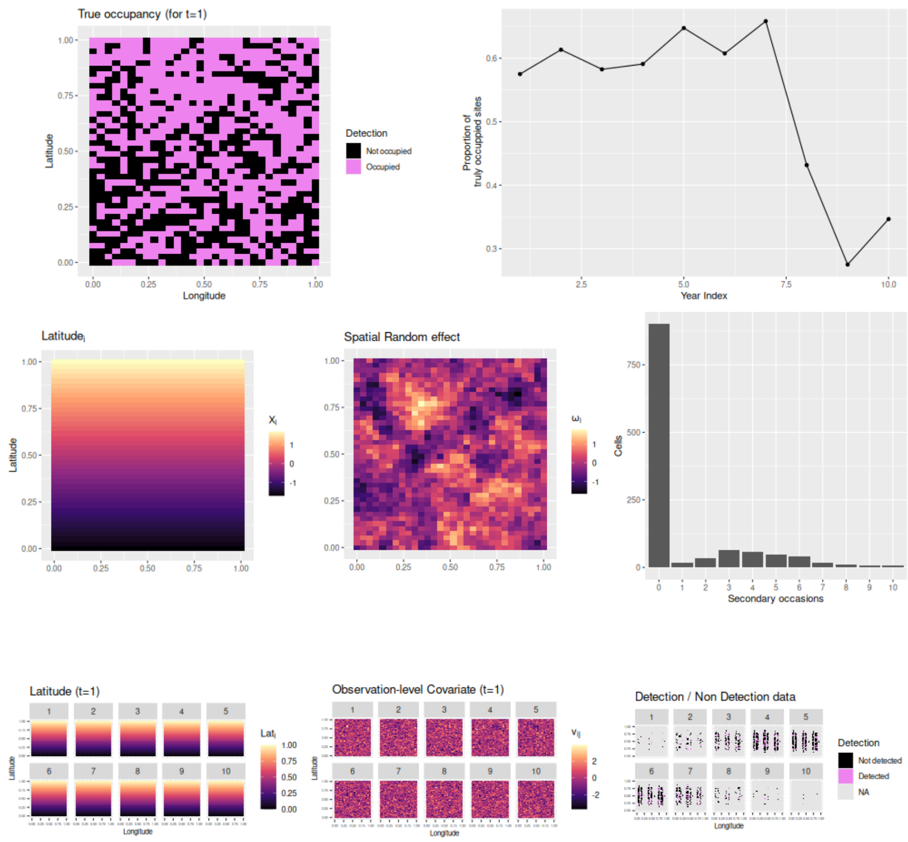

Figure A.10: Data design of our study 3 scenario 3 (3-3). The design was based on a Poisson distribution with average $\lambda = 1.1 \times (I/I^*)$ to spread J secondary occasions across sites and primary occasions (middle row, right), latitude L_i and v_{itj} as detection predictors, phenology and spatial spot grouping observations in space and time (bottom row). Note that spatial gaps are generated in this scenario. The subscripts t were omitted because only data simulated for $t = 1$ are shown.

B Supporting Information B - Simulation results

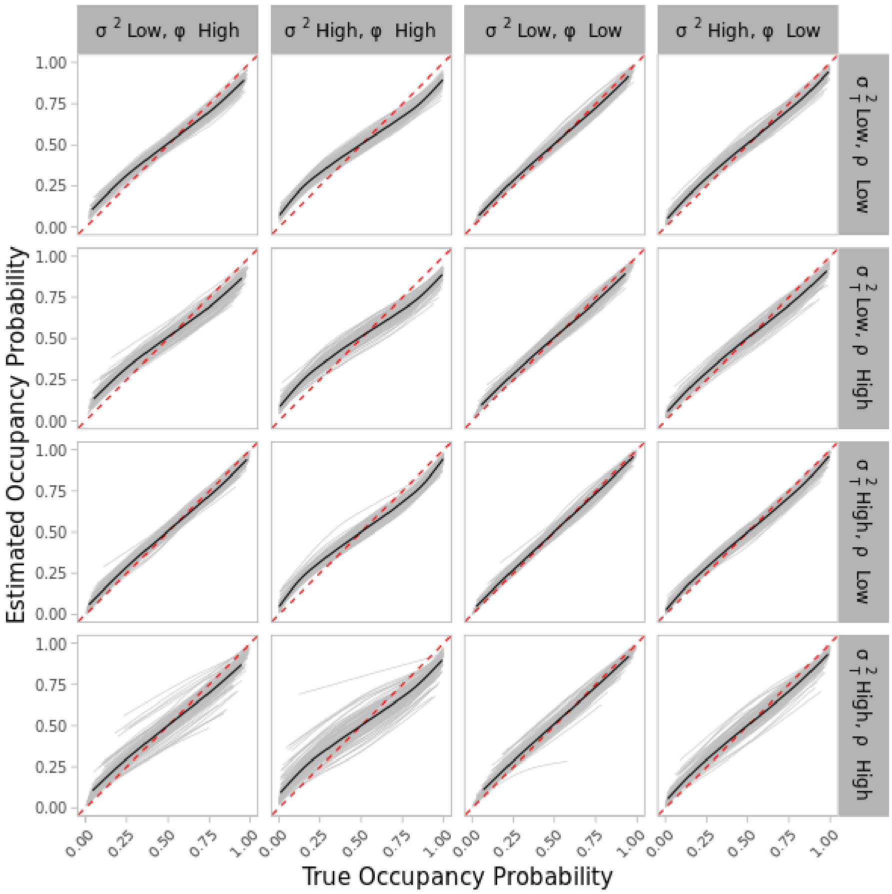

Figure B.1: Relationship between the true ψ_{it} (x-axis) and the estimated $\hat{\psi}_{it}$ (y-axis) occupancy probability for the replication of original simulations of D&S (study 1, scenario 0). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated data set. The black line depicts the averaged relationship across 100 simulated data sets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

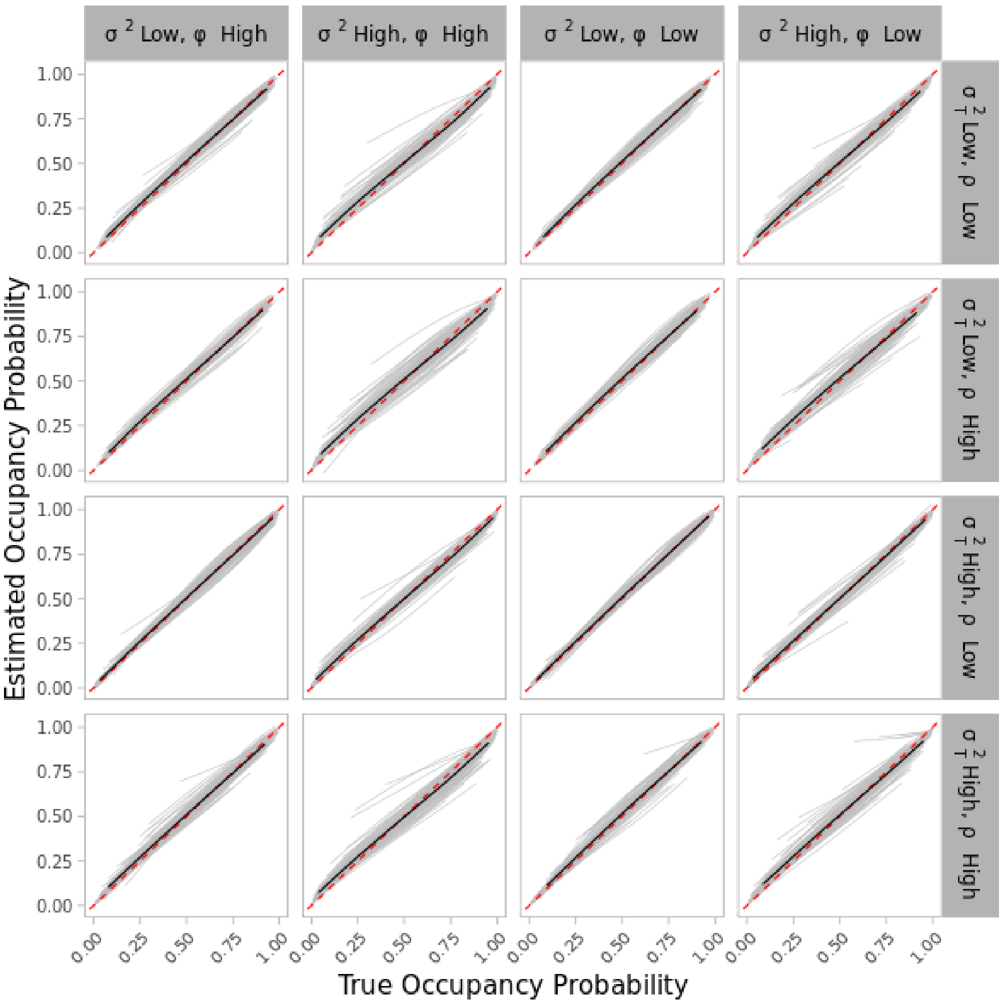

Figure B.2: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis) when spatial autocorrelation was high ($\phi = 0.5$ and $\phi = 1$) (study 1-scenario 1). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated data set. The black line depicts the averaged relationship across the 100 simulated data sets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

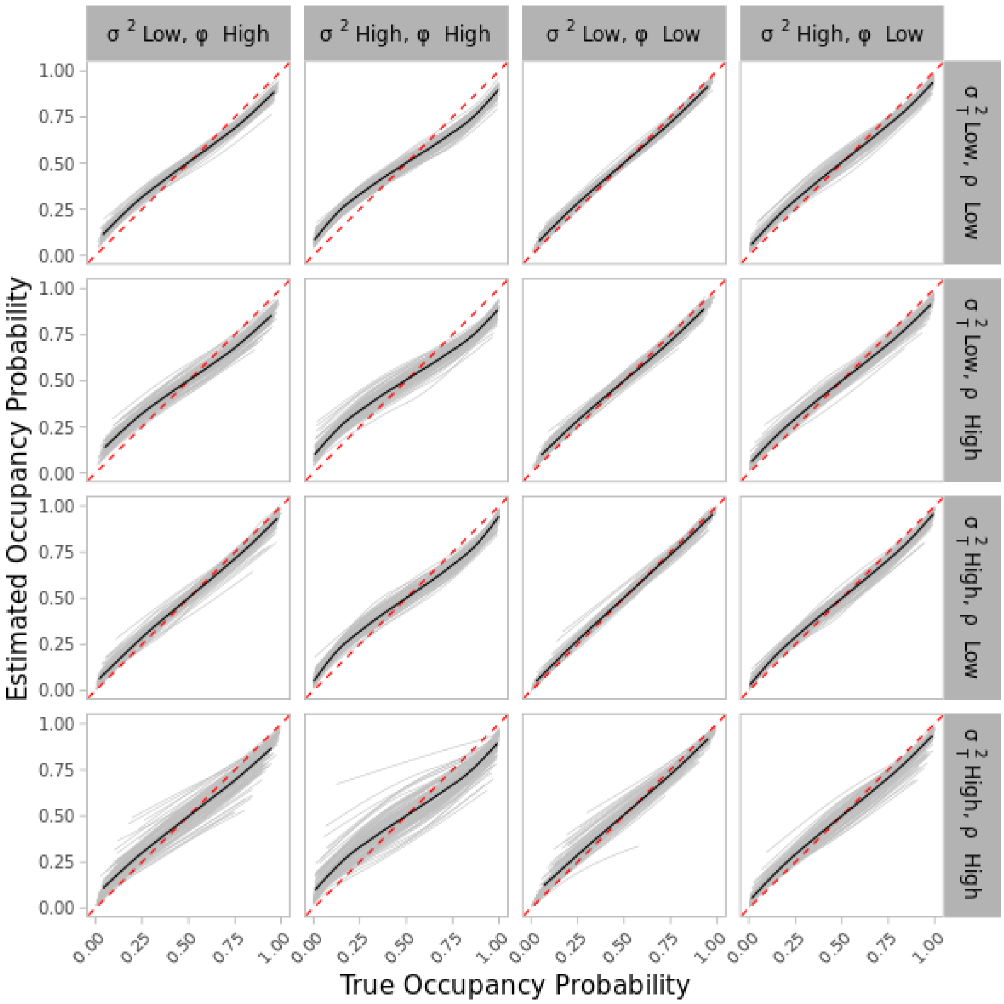

Figure B.3: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi_{it}}$ (y-axis) when shifting the sampling design from Bernoulli to a Poisson distribution of the number of surveys (study 2-scenario 0). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi_{it}}$ per simulated data set. The black line depicts the averaged relationship across the 100 simulated data sets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

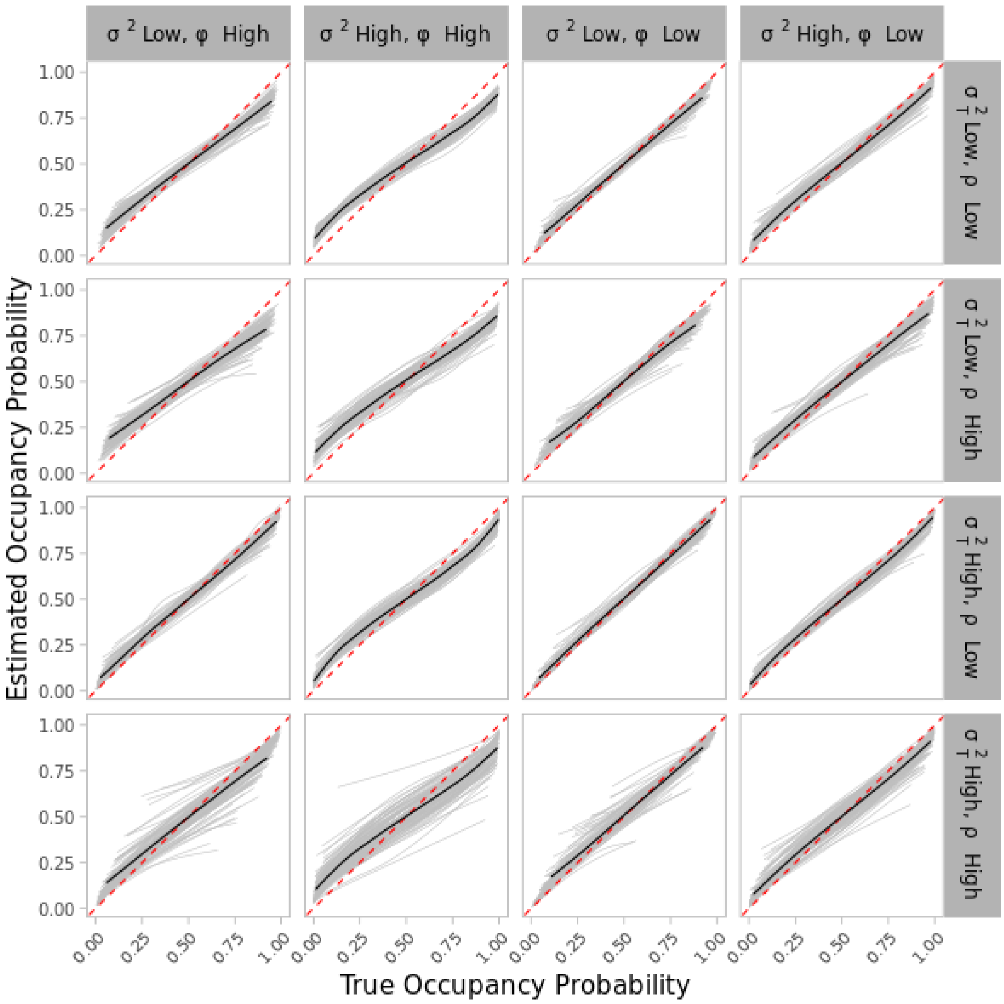

Figure B.4: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis) when X_{it} was replaced by a temporally constant covariate L_i in the occupancy model (study 2-scenario1). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated data set. The black line depicts the averaged relationship across the 100 simulated data sets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

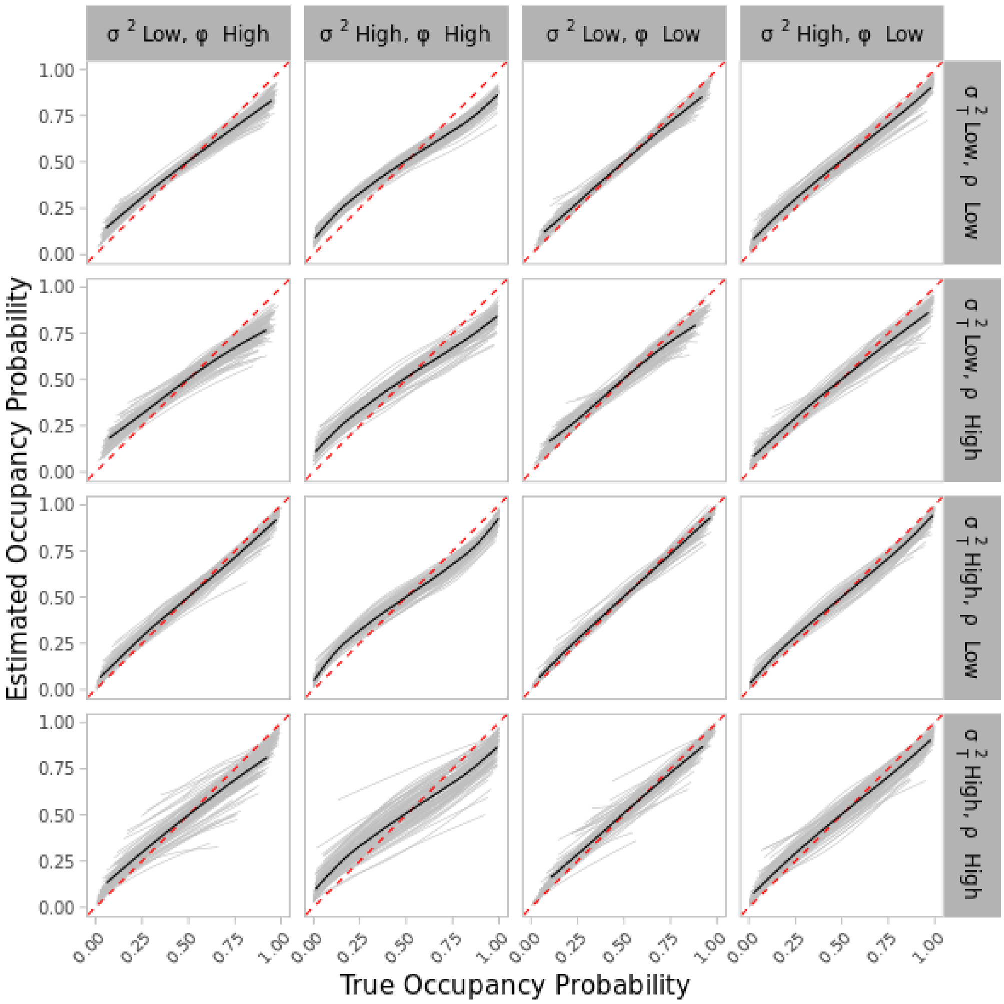

Figure B.5: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis) when X_{it} was replaced by L_i in the occupancy model, and the detection model had L_i and v_{itj} as predictors (study 2-scenario 3). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated data set. The black line depicts the averaged relationship across the 100 simulated data sets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

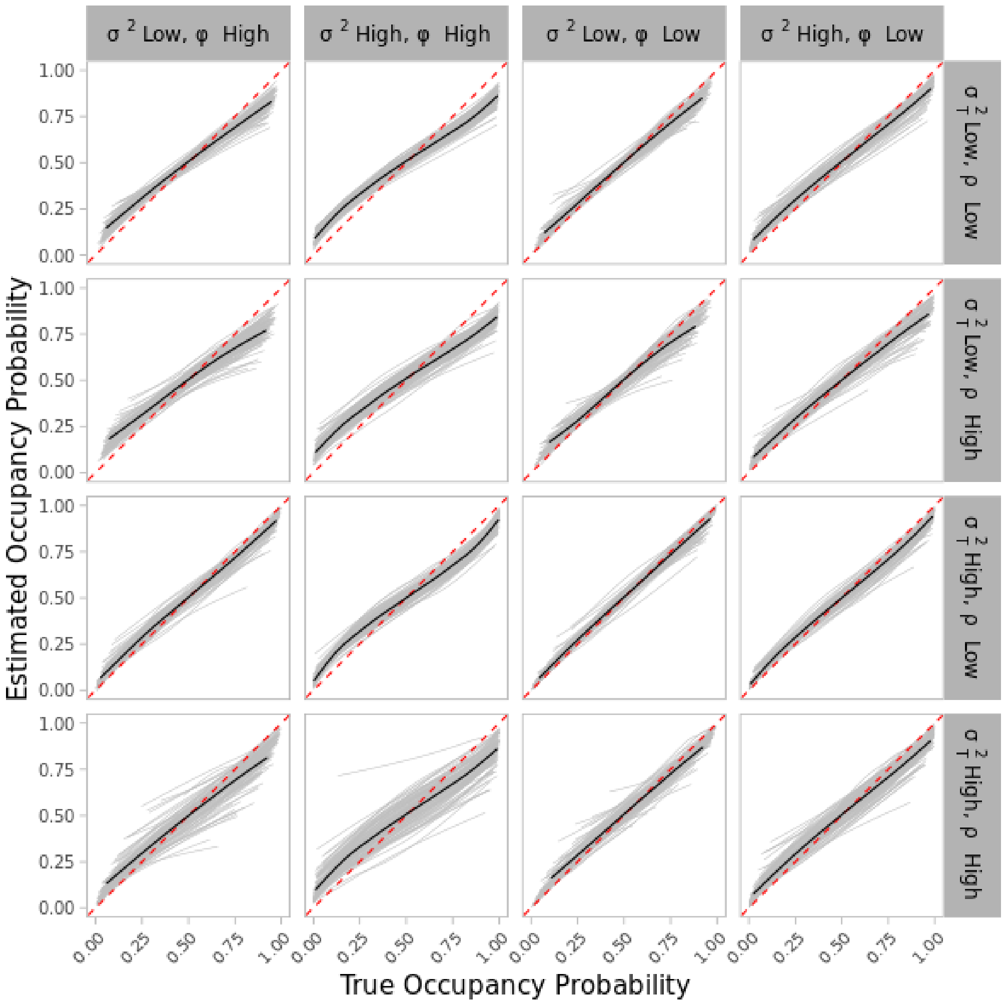

Figure B.6: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis) when a phenology effect was imposed to the occupancy data (study 3-scenario 1). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated data set. The black line depicts the averaged relationship across the 100 simulated data sets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

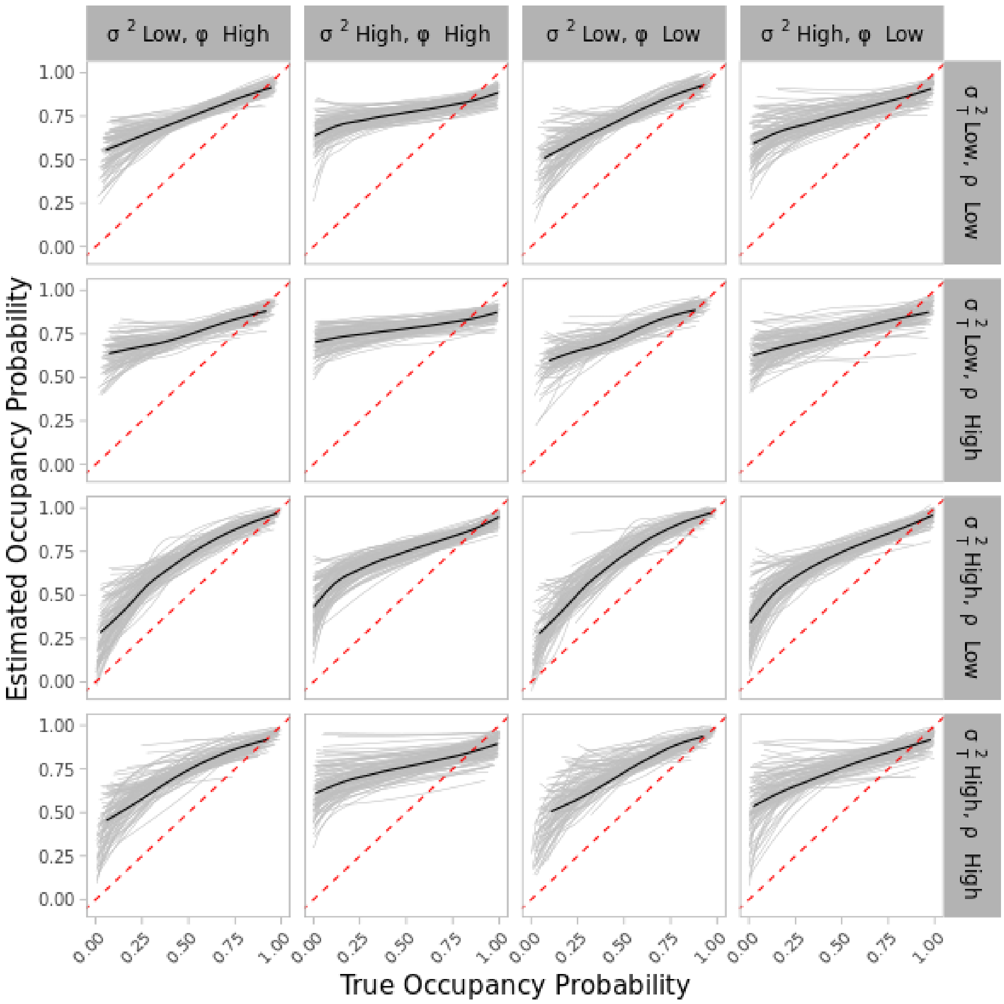

Figure B.7: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis) for study 3-scenario 2 (spatial and temporal clusters of observations, lower amount of data). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated data set. The black line depicts the averaged relationship across the 100 simulated data sets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

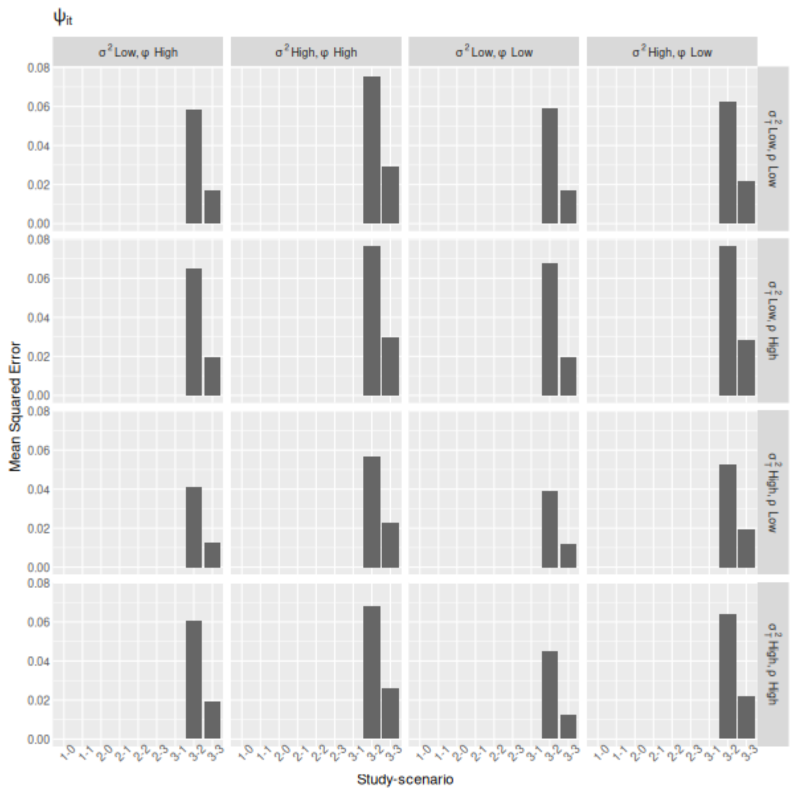

Figure B.8: Mean Squared Error of ψ_{it} estimator across studies, scenarios, and sub scenarios of spatial and temporal autocorrelation (facet labels).

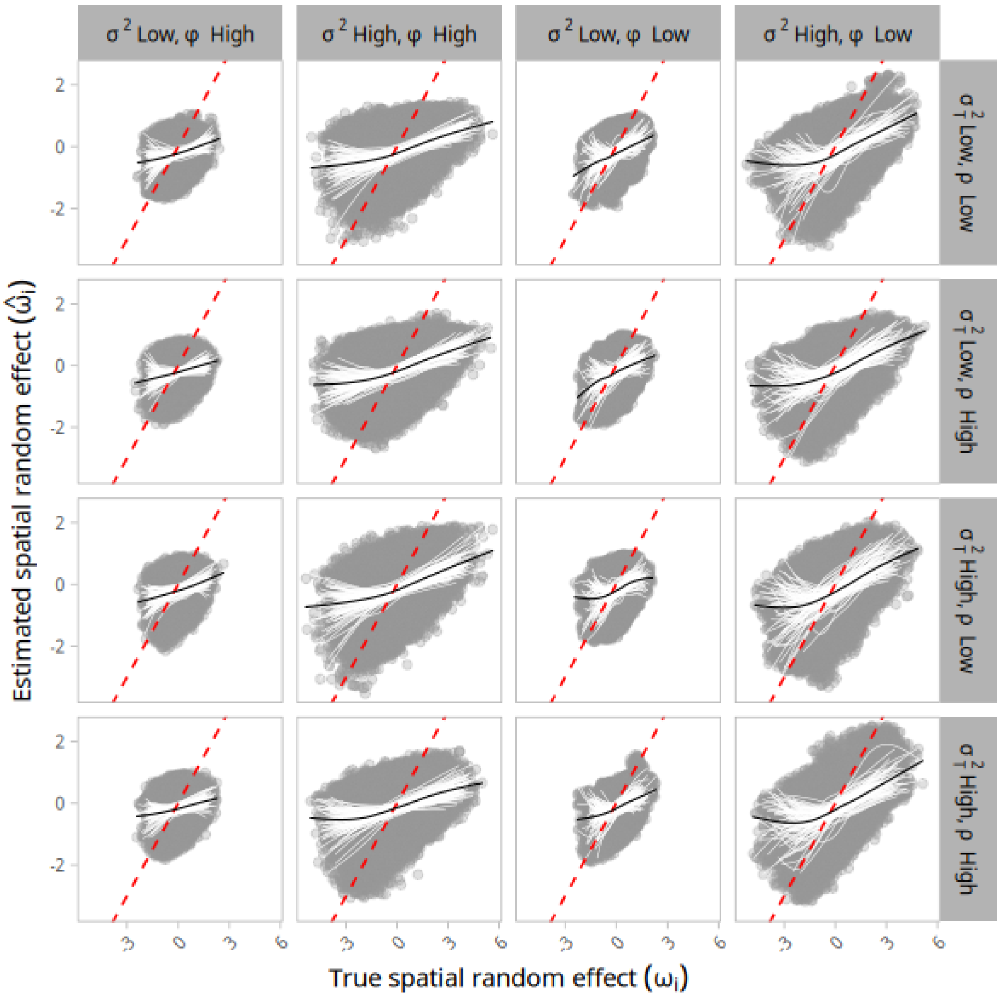

Figure B.9: Relationship between the true spatial random effect ω_i (x-axis) and the estimated spatial random effect $\hat{\omega}_i$ (y-axis) when phenology and observation spot were imposed to occupancy data (study 3 - scenario 3). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ω_i and $\hat{\omega}_i$ per simulated data set. The black line depicts the averaged relationship across the 100 simulated data sets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

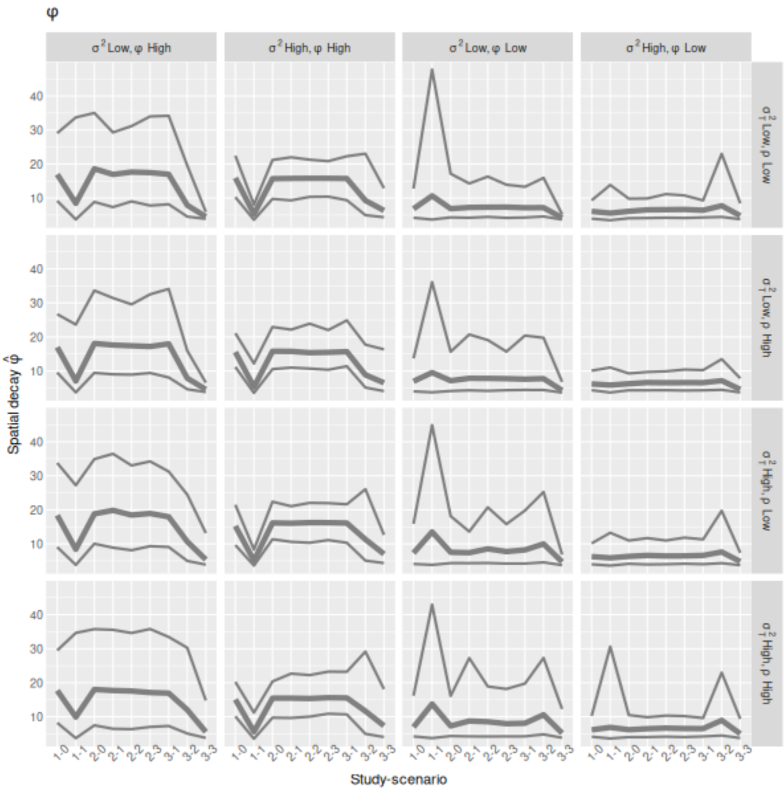

Figure B.10: Point estimates and 95% Credible Intervals (upper and lower bounds) around estimates of the spatial decay parameter $\hat{\phi}$ across studies and scenarios (x-axis). Point estimates were calculated using the 3,000 posterior distribution draws produced by each simulated data set. Spatial and temporal autocorrelation sub scenarios are represented across columns and rows, respectively.

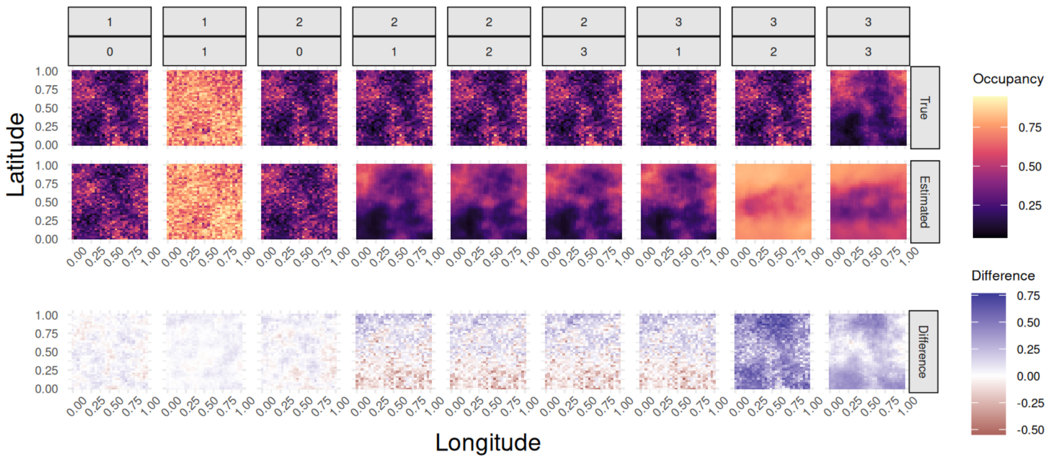

Figure B.11: Spatial map of true site occupancy probability ψ_{it} (top), estimated site occupancy probability $\hat{\psi}_{it}$ (middle), and the difference between them (bottom) across studies and scenarios (plot facets). Positive differences (in blue) indicate overestimation relative to the true occupancy, whereas negative differences (in red) indicate underestimate relative to the true occupancy. Results for the second simulated data set, and first primary period of parameter estimation ($t = 1$) with low autocorrelation parameter values ($\phi = 3.75$ (or $\phi = 0.5$ in study-scenario 1-1), $\sigma^2 = 0.3$, $\rho = 0.5$, $\sigma_T^2 = 0.3$).

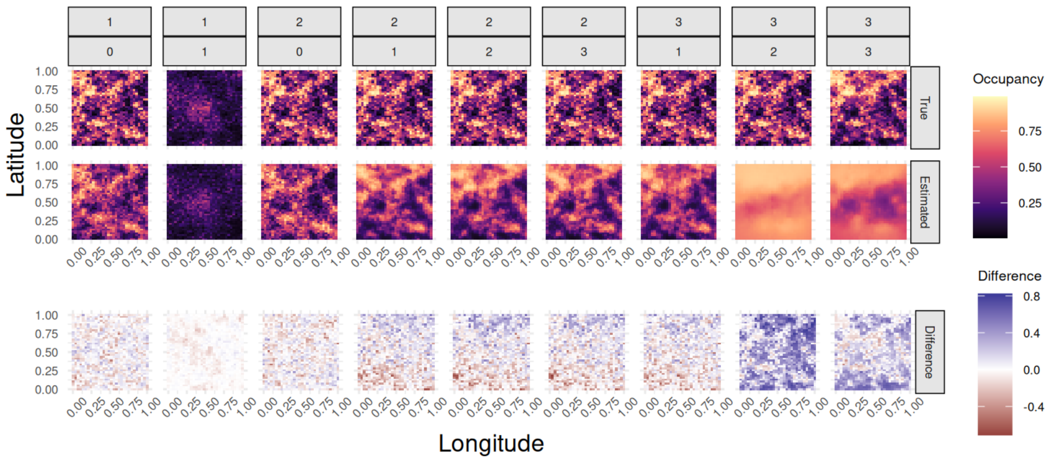

Figure B.12: Spatial map of true site occupancy probability ψ_{it} (top), estimated site occupancy probability $\hat{\psi}_{it}$ (middle), and the difference between them (bottom) across studies and scenarios (plot facets). Positive differences (in blue) indicate overestimation relative to the true occupancy, whereas negative differences (in red) indicate underestimate relative to the true occupancy. Results for the first simulated data set and the first primary period of parameter estimation ($t = 1$) with high autocorrelation parameter values ($\phi = 15$ (or $\phi = 1$ in study-scenario 1-1), $\sigma^2 = 1.5$, $\rho = 0.9$, $\sigma_T^2 = 1.5$).

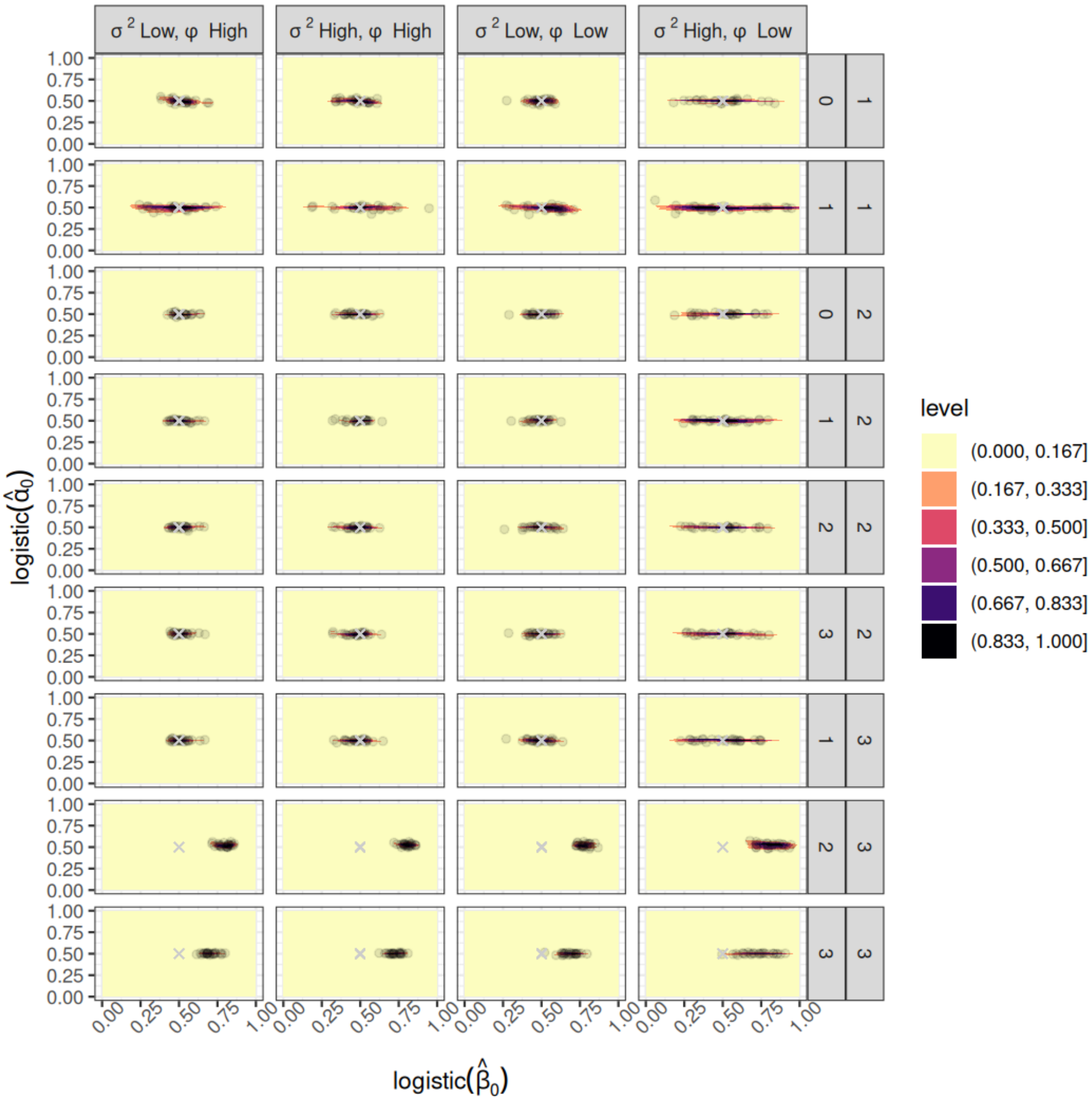

Figure B.13: Contour plots showing the combination of point estimates of occupancy ($\hat{\beta}_0$, x-axis) and detection intercepts ($\hat{\alpha_0}$, y-axis). Both quantities are shown in the probability scale (inverse logit). Studies and data design scenarios are shown across the facet rows, and spatial autocorrelation sub scenarios shown along facet columns. These results were produced by the sub scenarios with low temporal correlation $\rho = 0.5$ and temporal variance $\sigma_T^2 = 0.3$. The density is given by 100 simulation runs per data set. 'level' in the legend depicts the discretized proportion of runs within each contour plot region. The gray crosses depict the true parameter values, where α_0 and β_0 were both 0 at logistical/logit scale, and 0.5 at the probability scale (showed in the plot).

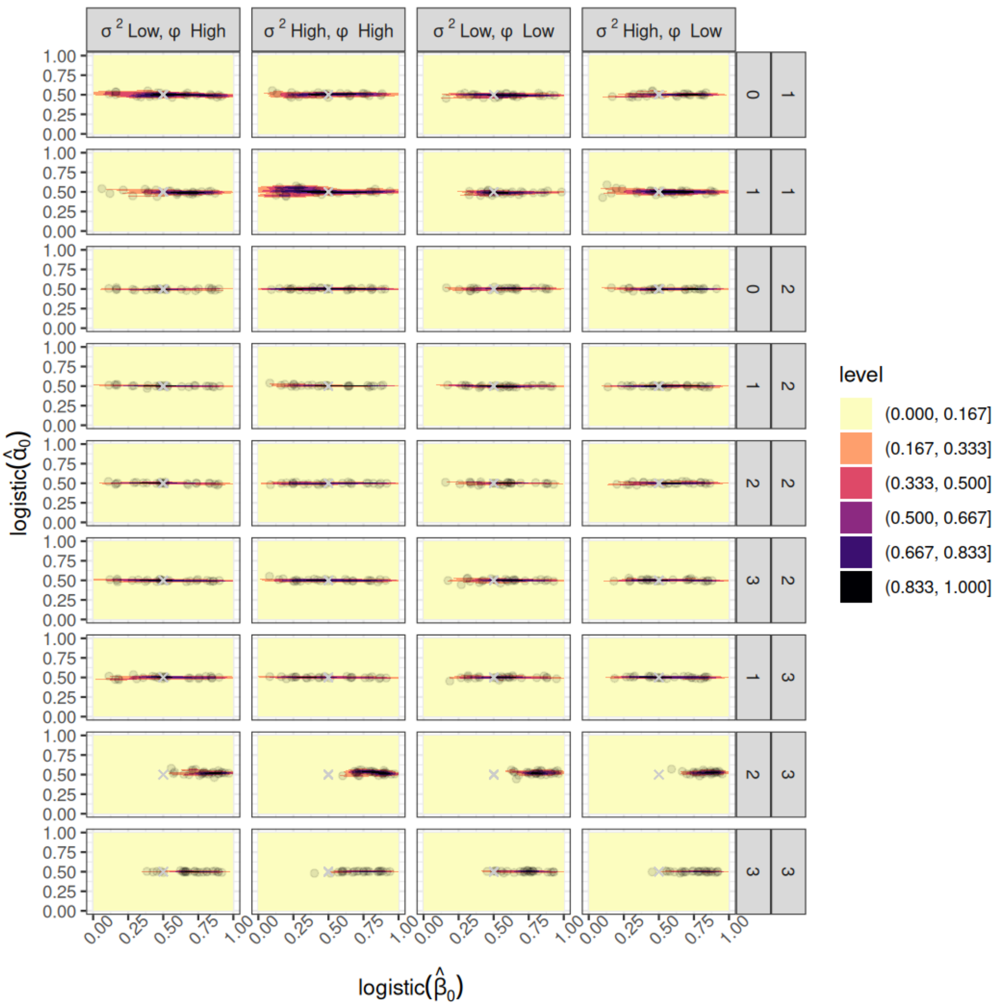

Figure B.14: Contour plots showing the combination of point estimates of occupancy ($\hat{\beta}_0$, x-axis) and detection intercepts ($\hat{\alpha}_0$, y-axis). Both quantities are shown in the probability scale (inverse logit). Studies and data design scenarios are shown across the facet rows, and spatial autocorrelation sub scenarios shown along facet columns. These results were produced by the sub scenarios with high temporal correlation $\rho = 0.9$ and temporal variance $\sigma_T^2 = 1.5$. The density is given by 100 simulation runs per data set. 'level' in the legend depicts the discretized proportion of runs within each contour plot region. The gray crosses depict the true parameter values, where α_0 and β_0 were both 0 at logistical/logit scale, and 0.5 at the probability scale (showed in the plot).

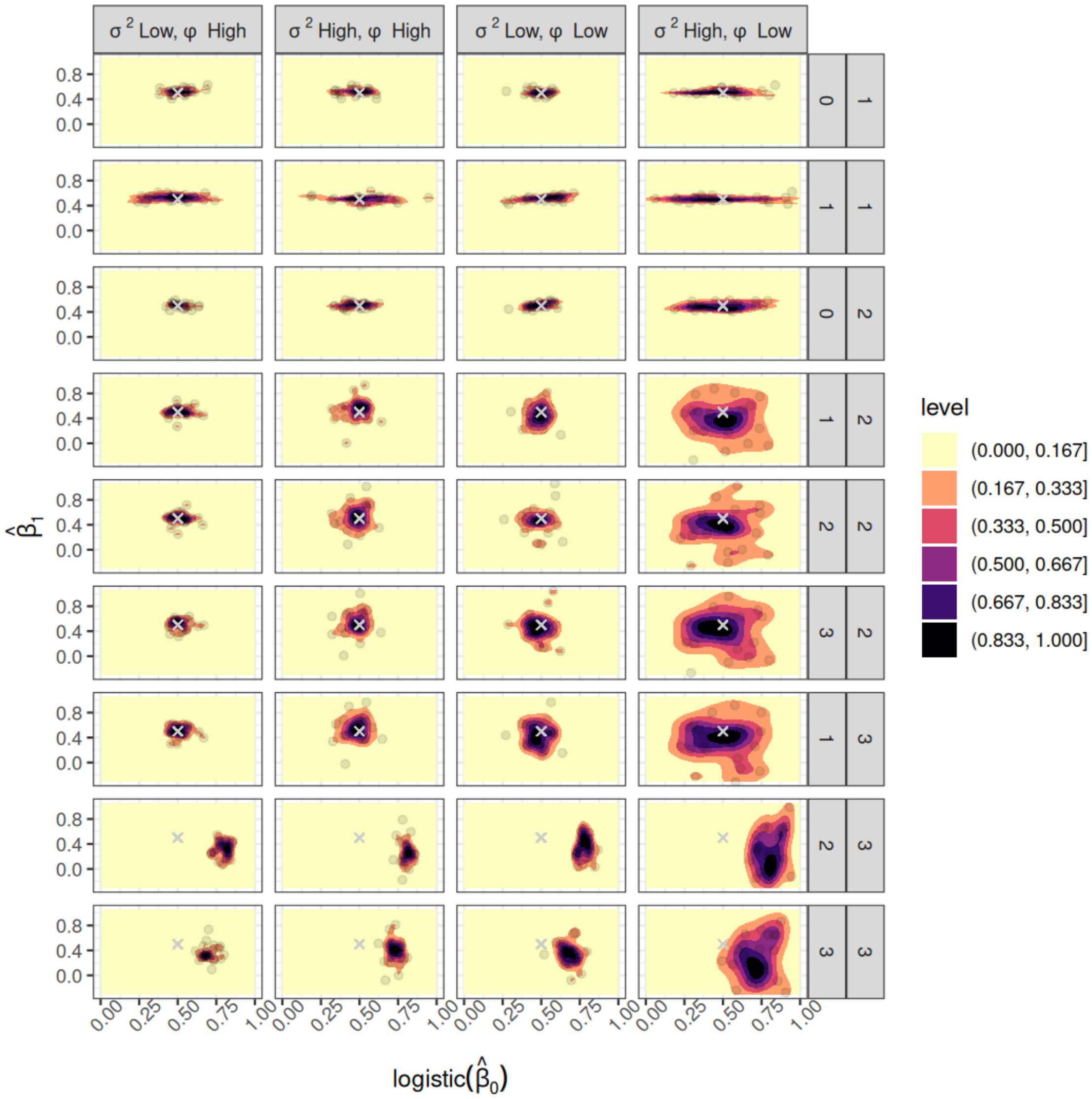

Figure B.15: Contour plots showing the combination of point estimates of occupancy intercept ($\hat{\beta}_0$, x-axis) and slope ($\hat{\beta}_1$, y-axis). The intercept is shown in the probability scale (inverse logit). $\hat{\beta}_1$ represents the effect of X_{it} on occupancy for study 1-scenario 0, study 1-scenario 1, and study 2-scenario 0; for the others, it represents the effect of latitude L_i on occupancy. Studies and data design scenarios are shown across the facet rows, and spatial autocorrelation sub scenarios shown along facet columns. These results were produced by the sub scenarios with low temporal correlation $\rho = 0.5$ and temporal variance $\sigma_T^2 = 0.3$. The density is given by 100 simulation runs per data set. 'level' in the legend depicts the discretized proportion of runs within each contour plot region. The gray crosses depict the true parameter values, where $\beta_0 = 0.5$ (probability scale) and $\beta_1 = 0.5$.

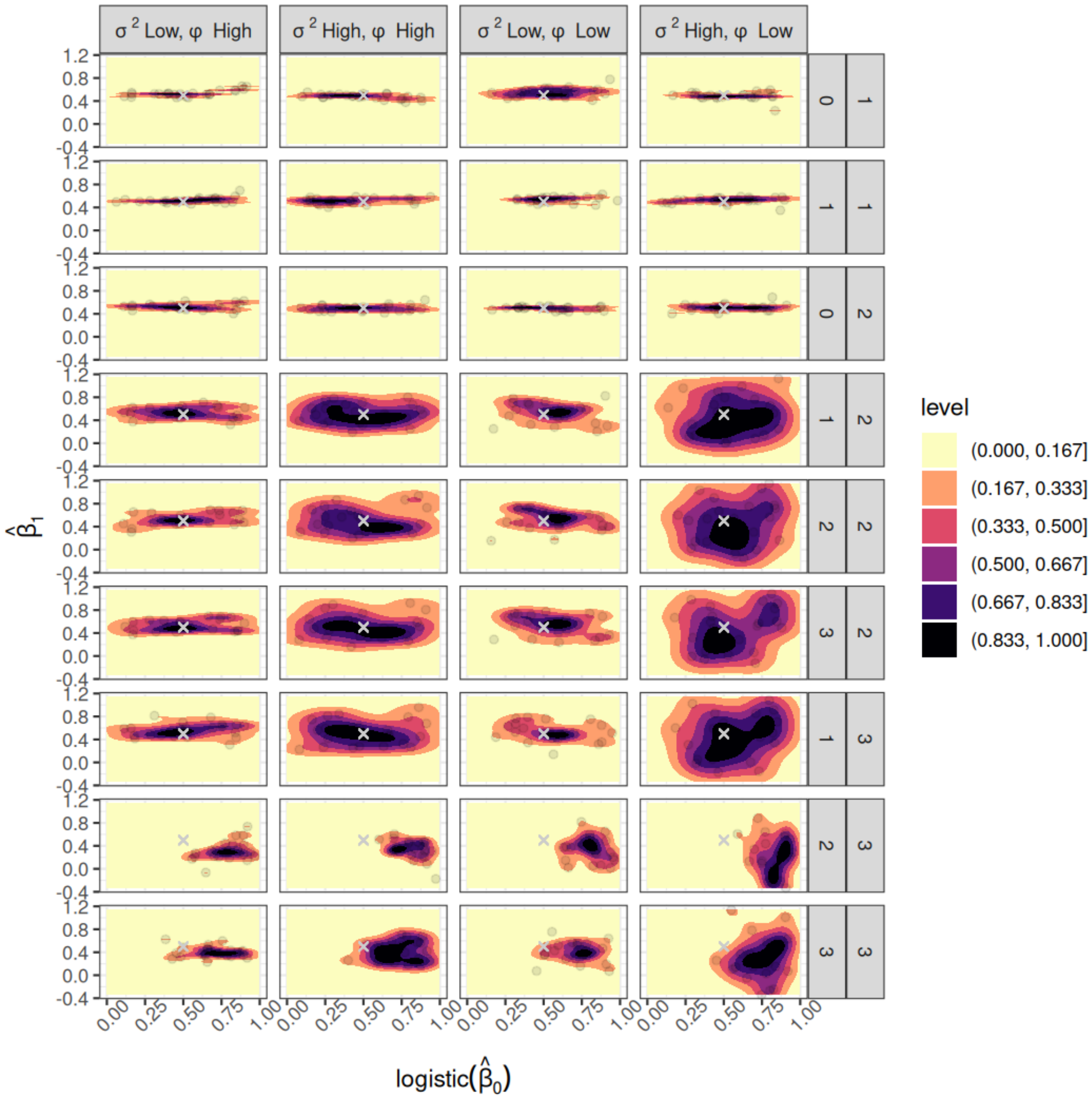

Figure B.16: Contour plots showing the combination of point estimates of occupancy intercept ($\hat{\beta}_0$, x-axis) and slope ($\hat{\beta}_1$, y-axis). The intercept is shown in the probability scale (inverse logit). $\hat{\beta}_1$ represents the effect of X_{it} on occupancy for study 1-scenario 0, study 1-scenario 1, and study 2-scenario 0; for the others, it represents the effect of latitude L_i on occupancy. Studies and data design scenarios are shown across the facet rows, and spatial autocorrelation sub scenarios shown along facet columns. These results were produced by the sub scenarios with high temporal correlation $\rho = 0.9$ and temporal variance $\sigma_T^2 = 1.5$. The density is given by 100 simulation runs per data set. 'level' in the legend depicts the discretized proportion of runs within each contour plot region. The gray crosses depict the true parameter values, where $\beta_0 = 0.5$ (probability scale) and $\beta_1 = 0.5$.

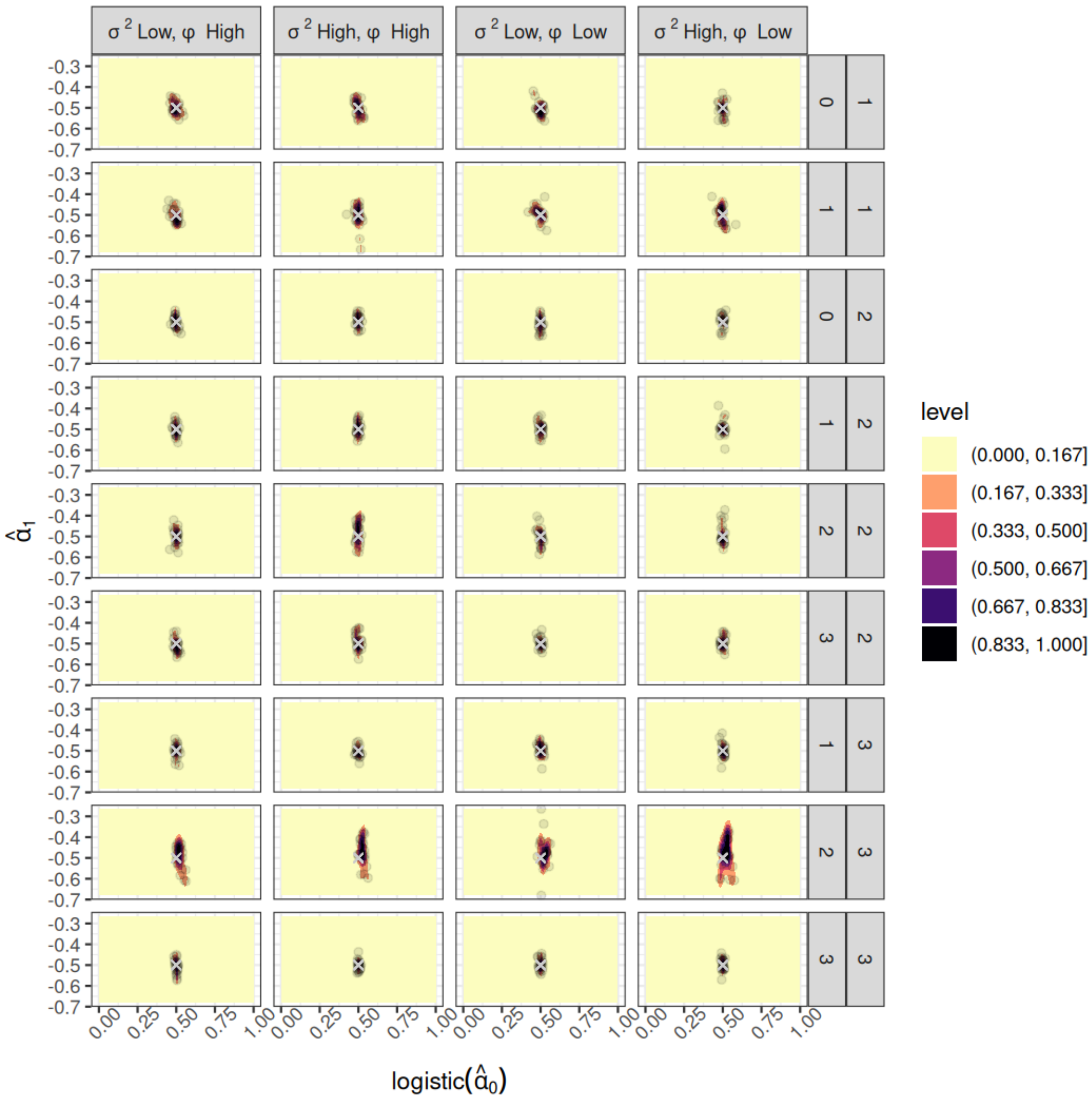

Figure B.17: Contour plots showing the combinations of point estimates of detection intercept ($\hat{\alpha_0}$, x-axis) and slope ($\hat{\alpha_1}$, y-axis). The intercept is shown in the probability scale (inverse logit). $\hat{\alpha_1}$ represents the effect of v_{itj} on detection for all studies-scenarios except for study 2-scenario 2, where $\hat{\alpha_1}$ represents the effect of latitude L_i on detection. Studies and data design scenarios are shown across the facet rows, and spatial autocorrelation sub scenarios shown along facet columns. These results were produced by the sub scenarios with low temporal correlation $\rho = 0.5$ and temporal variance $\sigma_T^2 = 0.3$. The density is given by 100 simulation runs per data set. 'level' in the legend depicts the discretized proportion of runs within each contour plot region. The gray crosses depict the true parameter values, where $\alpha_0 = 0.5$ (probability scale) and $\alpha_1 = -0.5$.

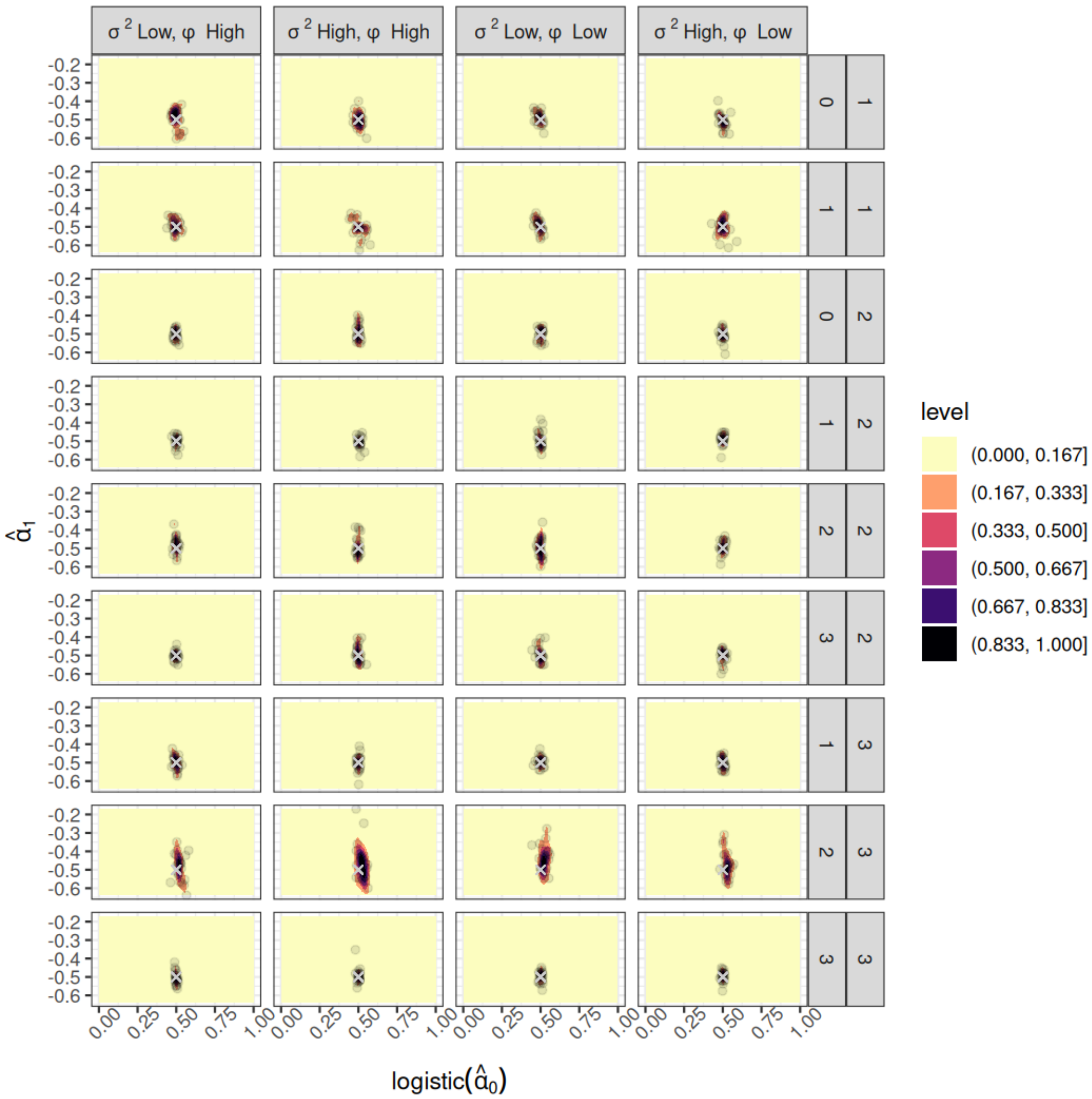

Figure B.18: Contour plots showing the combinations of point estimates of detection intercept ($\hat{\alpha_0}$, x-axis) and slope ($\hat{\alpha_1}$, y-axis). The intercept is shown in the probability scale (inverse logit). $\hat{\alpha_1}$ represents the effect of v_{itj} on detection for all studies-scenarios except for study 2-scenario 2, where $\hat{\alpha_1}$ represents the effect of latitude L_i on detection. Studies and data design scenarios are shown across the facet rows, and spatial autocorrelation sub scenarios shown along facet columns. These results were produced by the sub scenarios with high temporal correlation $\rho = 0.9$ and temporal variance $\sigma_T^2 = 1.5$. The density is given by 100 simulation runs per data set. 'level' in the legend depicts the discretized proportion of runs within each contour plot region. The gray crosses depict the true parameter values, where $\alpha_0 = 0.5$ (probability scale) and $\alpha_1 = -0.5$.

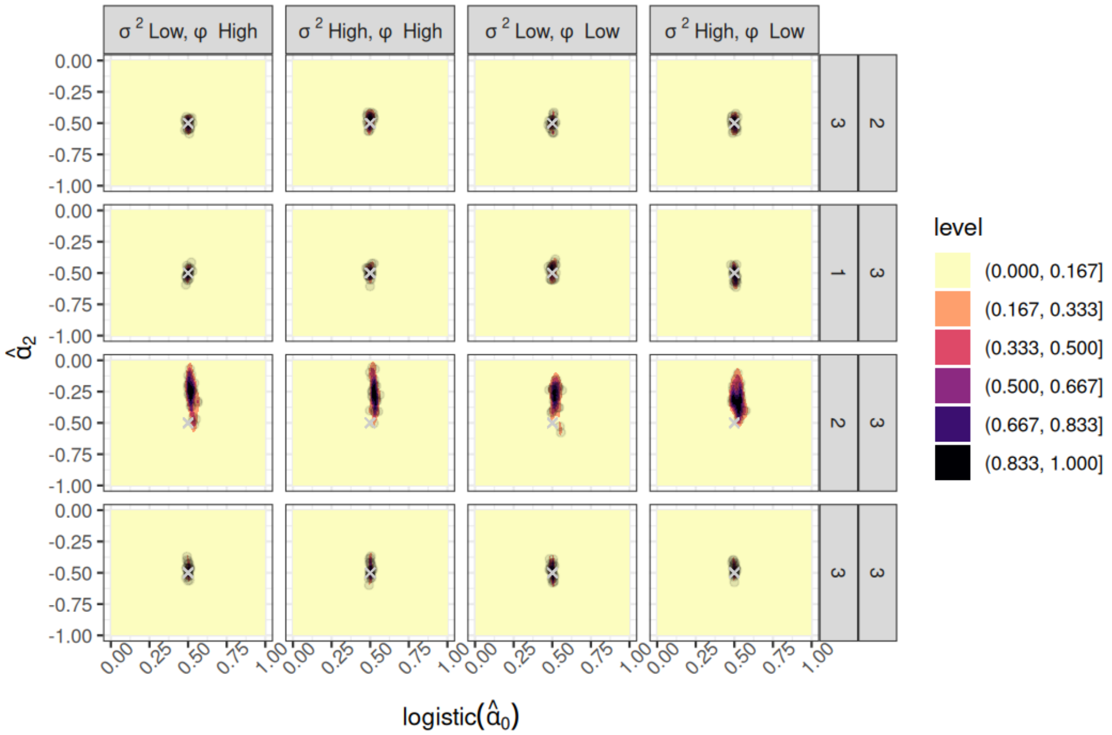

Figure B.19: Contour plots showing the combinations of point estimates of detection intercept ($\hat{\alpha_0}$, x-axis) and slope ($\hat{\alpha_2}$, y-axis). The intercept is shown in the probability scale (inverse logit). $\hat{\alpha_2}$ represents the effect of latitude L_i on detection probability. Studies and data design scenarios are shown across the facet rows, and spatial autocorrelation sub scenarios shown along facet columns. These results were produced by the sub scenarios with low temporal correlation $\rho = 0.5$ and temporal variance $\sigma_T^2 = 0.3$. The density is given by 100 simulation runs per data set. 'level' in the legend depicts the discretized proportion of runs within each contour plot region. The gray crosses depict the true parameter values, where $\alpha_0 = 0.5$ (probability scale) and $\alpha_2 = -0.5$.

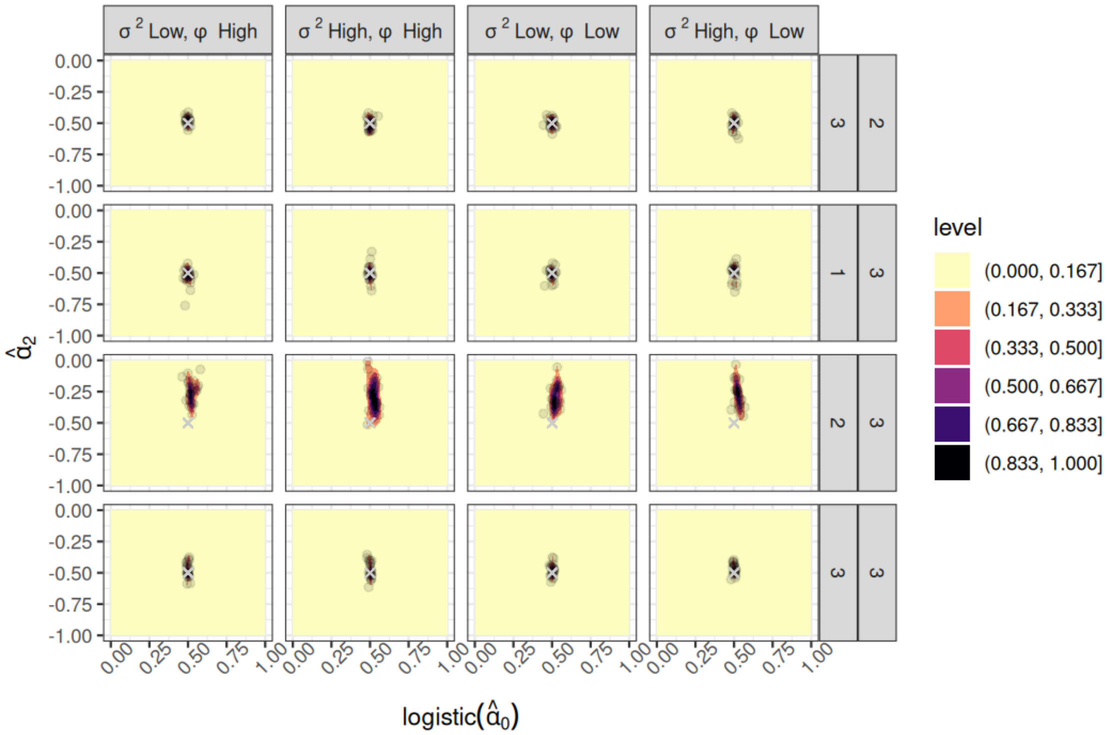

Figure B.20: Contour plots showing the combinations of point estimates of detection intercept ($\hat{\alpha_0}$, x-axis) and slope ($\hat{\alpha_2}$, y-axis). The intercept is shown in the probability scale (inverse logit). $\hat{\alpha_2}$ represents the effect of latitude L_i on detection probability. Studies and data design scenarios are shown across the facet rows, and spatial autocorrelation sub scenarios shown along facet columns. These results were produced by the sub scenarios with high temporal correlation $\rho = 0.9$ and temporal variance $\sigma_T^2 = 1.5$. The density is given by 100 simulation runs per data set. 'level' in the legend depicts the discretized proportion of runs within each contour plot region. The gray crosses depict the true parameter values, where $\alpha_0 = 0.5$ (probability scale) and $\alpha_2 = -0.5$.

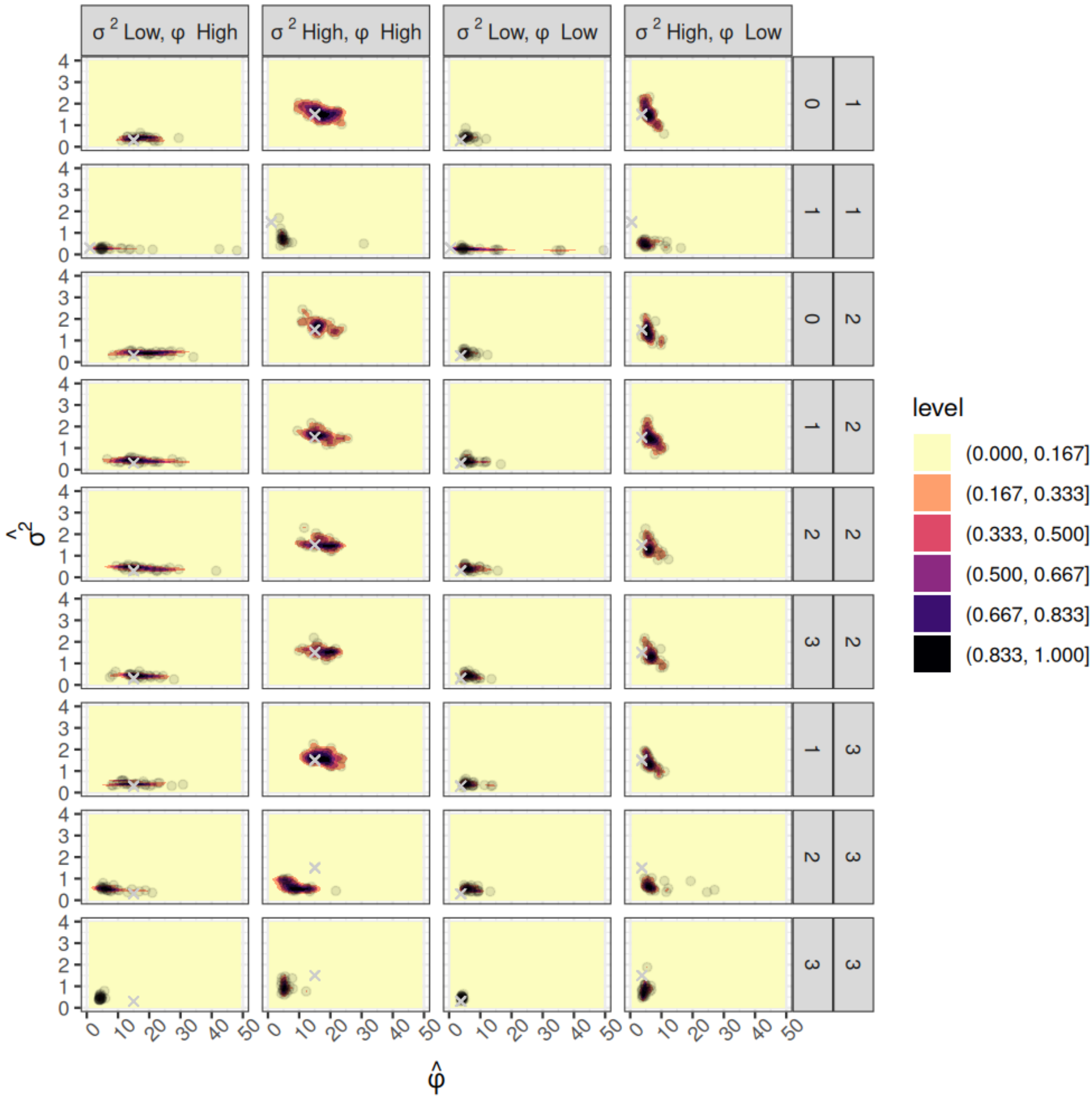

Figure B.21: Contour plots showing the combination of point estimates of the spatial decay ($\hat{\phi}$, x-axis) and variance ($\hat{\sigma^2}$, y-axis). Studies and data design scenarios are shown across the facet rows, and sub spatial autocorrelation scenarios shown along facet columns. These results were produced by the sub scenarios with high temporal correlation $\rho = 0.5$ and temporal variance $\sigma_T^2 = 0.3$. The density is given by 100 simulation runs per data set. 'level' in the legend depicts the discretized proportion of runs within each contour plot region. The gray crosses depict the true parameter values. Low $\phi = 3.75$ and high $\phi = 15$, except for study-scenario 1-1 where low $\phi = 0.5$ and high $\phi = 1$. Low $\sigma^2 = 0.3$ and high $\sigma^2 = 1.5$.

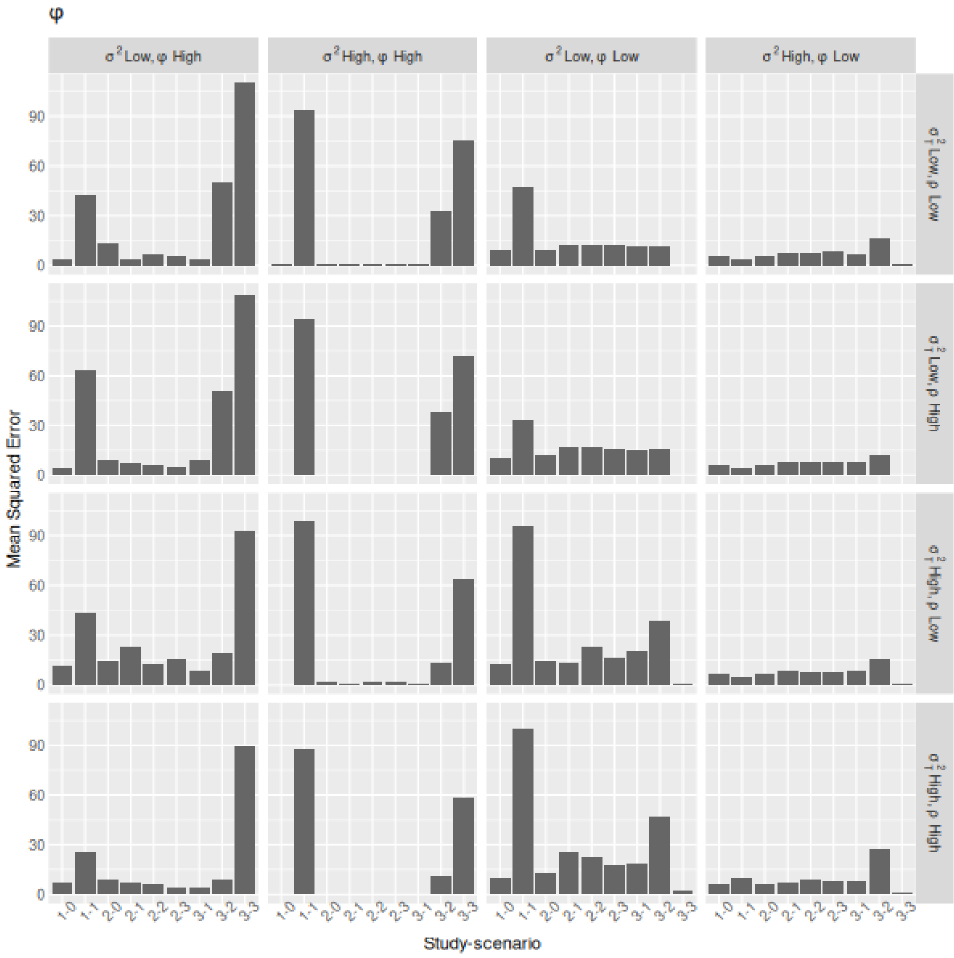

Figure B.22: Mean Squared Error of the spatial decay ϕ estimator across studies, scenarios, and sub scenarios of spatial and temporal autocorrelation (facet labels).

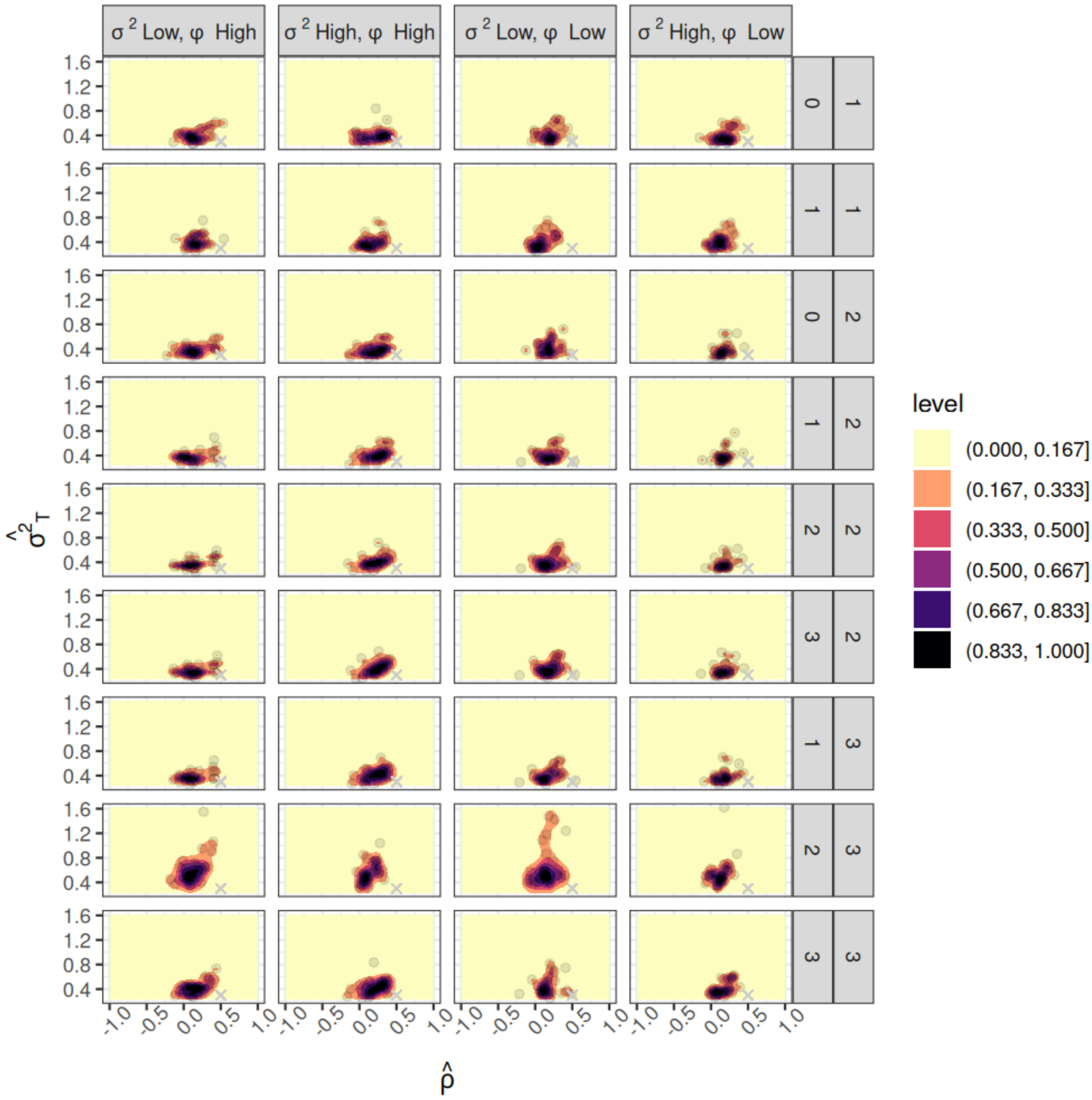

Figure B.23: Contour plots showing the combination of point estimates of the spatial decay ($\hat{\rho}$, x-axis) and variance ($\hat{\sigma_T^2}$, y-axis). Studies and data design scenarios are shown across the facet rows, and spatial autocorrelation sub scenarios shown along facet columns. These results were produced by the sub scenarios with low temporal correlation $\rho = 0.5$ and temporal variance $\sigma_T^2 = 0.3$ (gray crosses). The density is given by 100 simulation runs per data set. 'level' in the legend depicts the discretized proportion of runs within each contour plot region. The gray crosses depict the true parameter values.

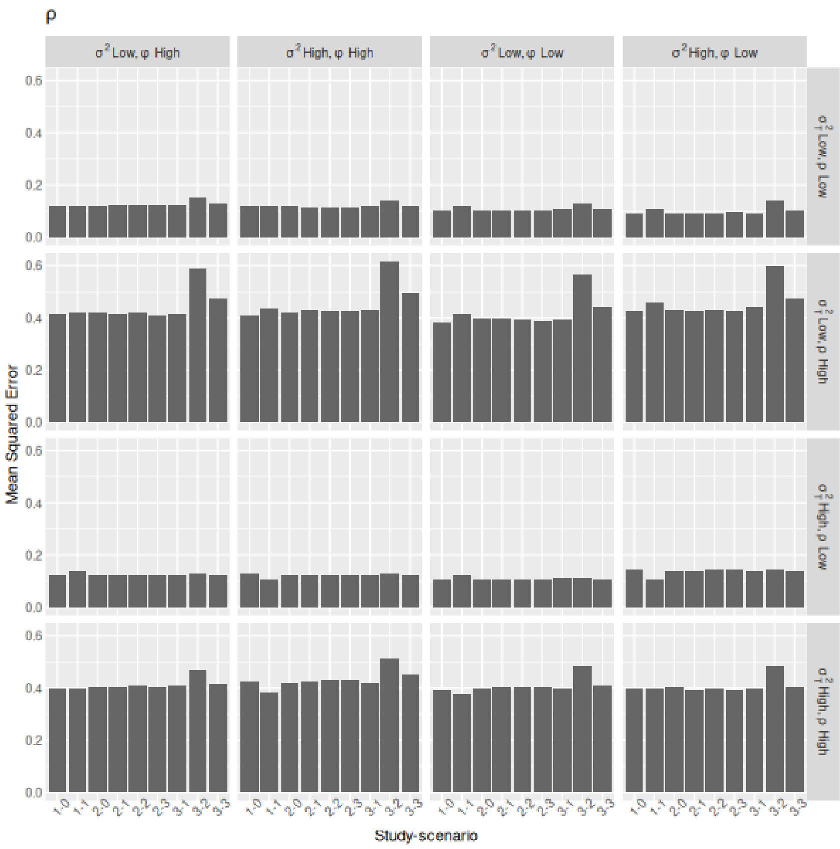

Figure B.24: Mean Squared Error of the temporal autocorrelation ρ estimator across studies, scenarios, and sub scenarios of spatial and temporal autocorrelation (facet labels).

C Supporting Information C - Results of the analyses using the empirical butterfly data

In this section we describe the covariates used in the models and show model results (Nouvelle Aquitaine and buffer scales). For the common blue, the scaled values of cell latitude and non-water land cover (linear and quadratic effects), and longitude, elevation and urban cover (linear effects) were set as occupancy predictors. For the large copper, we used latitude and marsh cover (linear and quadratic effects), and longitude, elevation, urban cover (linear effects) as occupancy predictor to represent habitat affinities (Gourvil & Sannier, 2020). Scaled cell latitude, number of observers (count of unique observer IDs associated to the aggregated records, where one ID could be of a single individual or group), non-water cover (linear effects) and survey month (linear and quadratic effects) were used as detection predictors for both species. Elevation was obtained from the European Digital Elevation Model (EU-DEM, Copernicus data at 10 m resolution, downloaded on 2024-04-27), and land cover data were gathered from the CORINE habitat classification scheme (reference year: 2018) at 100 m resolution, downloaded on 2024-06-04. Urban cover included the cell area with continuous urban/fabric structures. Marsh cover/humid areas included the sum of the cover of inland marshes, peat bogs, salt marshes, water courses and bodies, coastal lagoons and estuaries. Elevation and habitat data were averaged at 1×1 km cell scale. Latitude, longitude, number of observers and survey month were extracted from the butterfly dataset.

Table C.1: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to the common blue *P. icarus* occupancy data (buffer data). Results for models with 15 spatial neighbors, and weak or informative ϕ priors. PPO: Prior-Posterior Overlap. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, LAND= non-water land cover, URBAN = continuous urban/fabric cover, OBS: Number of observers, MON: survey month.

-	Weak prior, NNGP=15					Informative prior, NNGP=15				
Parameter	Mean	LCI	UCI	Rhat	Overlap	Mean	LCI	UCI	Rhat	Overlap
β_0	0.237	-0.951	1.459	1.039	57.5	0.316	-0.724	1.364	1.215	50.0
β_{LAT}	-0.024	-0.655	0.603	1.143	37.3	0.019	-0.505	0.604	1.071	31.5
β_{LAT^2}	0.070	-0.394	0.560	1.085	30.4	0.047	-0.449	0.532	1.009	28.3
β_{LON}	0.086	-0.360	0.614	1.001	29.9	0.076	-0.364	0.524	1.001	26.3
β_{ELEV}	-0.376	-0.978	0.102	1.007	30.8	-0.339	-0.815	0.101	1.002	26.8
$\beta_{NON-WATER}$	-1.218	-3.352	0.840	1.026	61.8	-1.260	-3.377	0.539	1.021	58.7
$\beta_{NON-WATER^2}$	-0.115	-1.014	1.631	1.869	50.4	-0.255	-1.087	0.543	1.035	41.9
β_{URBAN}	-0.647	-1.073	-0.288	1.021	22.2	-0.632	-1.141	-0.316	1.039	23.6
σ^2	4.862	1.389	9.879	1.215	25.6	4.251	1.522	12.104	1.295	29.5
ϕ	32.491	4.555	58.696	1.014	8.7	1.720	0.560	2.913	1.002	93.0
σ^2_T	1.918	0.581	4.830	1.060	46.8	1.737	0.517	4.926	1.110	47.2
ρ	0.076	-0.251	0.399	1.005	39.7	0.087	-0.239	0.403	0.999	40.8
α_0	-0.159	-0.502	0.177	1.014	20.8	-0.148	-0.489	0.203	1.011	21.5
α_{LAT}	-0.228	-0.419	-0.040	1.078	12.4	-0.245	-0.433	-0.063	1.043	12.1
α_{OBS}	0.027	-0.001	0.056	1.000	2.3	0.027	-0.001	0.056	1.006	2.4
α_{MON}	0.890	0.679	1.116	1.019	13.3	0.896	0.676	1.112	1.004	13.2
α_{MON^2}	-1.692	-2.013	-1.403	1.003	11.8	-1.706	-2.011	-1.405	1.004	11.8
$\alpha_{NON-WATER}$	0.111	-0.032	0.264	1.255	9.9	0.101	-0.037	0.233	1.111	10.0

Table C.2: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to the large copper *L. dispar* occupancy data (buffer data). Results for models with 15 spatial neighbors, and weak or informative ϕ priors. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, MAR=natural marshes/humid areas, LAND= non-water land cover, URBAN = continuous urban/fabric cover, OBS: Number of observers, MON: survey month.

-	Weak prior, NNGP=15					Informative prior, NNGP=15				
Parameter	Mean	LCI	UCI	Rhat	Overlap	Mean	LCI	UCI	Rhat	Overlap
β_0	-4.177	-5.488	-2.918	1.010	6.3	-4.255	-5.556	-2.932	1.276	7.6
β_{LAT}	0.121	-0.549	0.781	1.067	35.8	0.145	-0.456	0.861	1.024	35.1
β_{LAT^2}	0.301	-0.215	0.839	1.014	30.3	0.312	-0.251	0.879	1.014	30.3
β_{LON}	1.590	0.693	2.570	1.110	32.3	1.678	0.651	2.769	1.319	33.4
β_{ELEV}	-2.398	-3.177	-1.700	1.084	16.4	-2.425	-3.206	-1.663	1.029	18.6
$\beta_{MARSHES}$	0.565	-0.427	1.769	1.007	50.4	0.635	-0.433	1.888	1.085	49.9
$\beta_{MARSHES^2}$	-0.450	-0.984	-0.074	1.011	25.8	-0.481	-1.025	-0.067	1.120	27.7
β_{URBAN}	-1.844	-4.450	-0.576	1.498	40.4	-1.903	-3.547	-0.710	1.211	38.2
σ^2	2.076	0.573	5.225	1.273	59.2	2.370	0.656	5.524	1.386	50.6
ϕ	31.629	4.405	57.933	1.000	9.0	1.751	0.538	2.936	1.007	97.3
σ_T^2	2.339	0.647	6.197	1.137	38.9	2.388	0.695	6.679	1.091	35.3
ρ	0.073	-0.261	0.400	1.024	43.2	0.070	-0.273	0.411	1.080	40.5
α_0	-1.480	-2.354	-0.636	1.014	32.0	-1.481	-2.246	-0.653	1.007	29.3
α_{LAT}	-0.399	-0.747	-0.088	1.015	21.0	-0.402	-0.715	-0.086	1.012	20.1
α_{OBS}	0.143	0.070	0.226	1.012	5.8	0.142	0.074	0.216	1.008	5.8
α_{MON}	0.250	-0.224	0.729	1.002	27.8	0.264	-0.219	0.707	1.000	28.7
α_{MON^2}	-1.865	-2.584	-1.231	1.001	21.9	-1.859	-2.647	-1.218	1.008	20.4
$\alpha_{NON-WATER}$	-0.194	-0.465	0.074	1.003	17.6	-0.189	-0.428	0.076	1.014	16.8

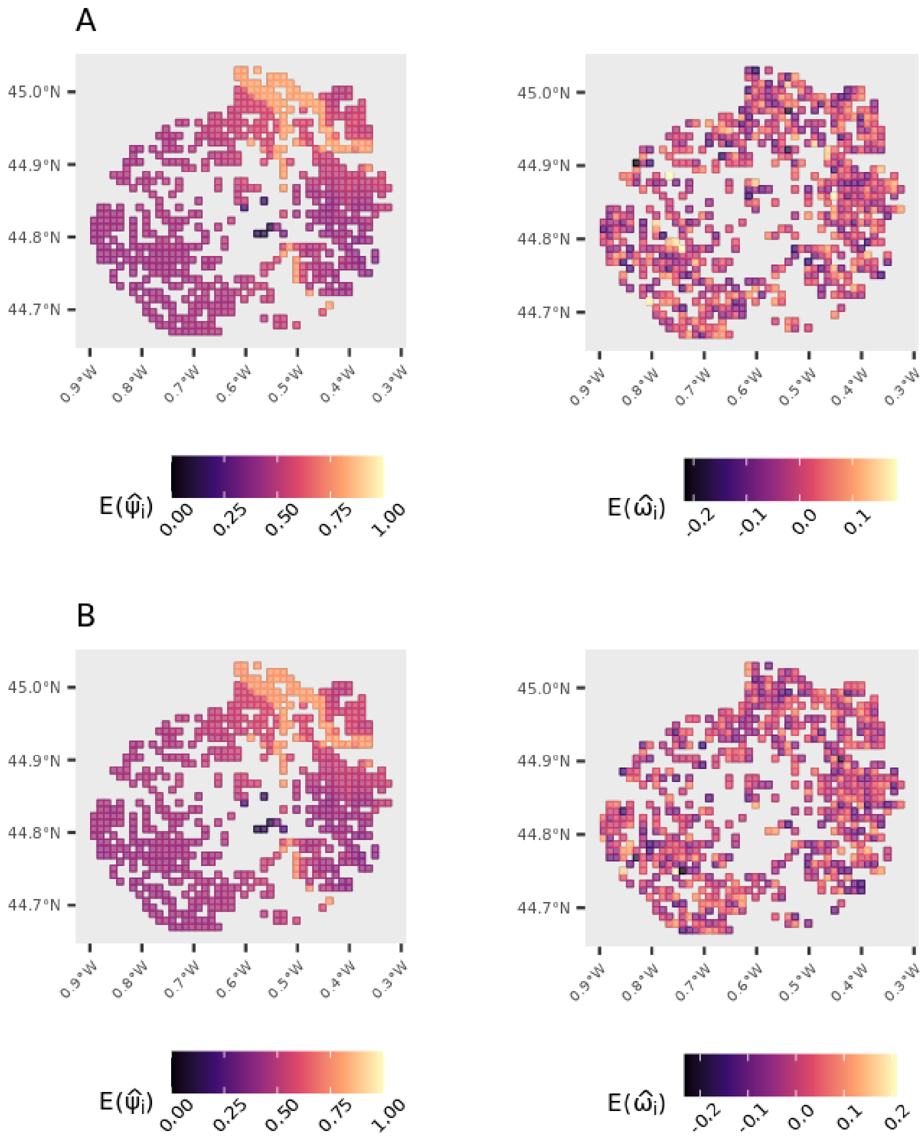

Figure C.1: Predicted occupancy $E(\hat{\psi}_i)$ of the common blue *P. icarus*, and the averaged values of the spatial random effect $E(\hat{\omega}_i)$ among the non-sampled cells in a buffer of 10 km around Bordeaux. Values for cells with butterfly records were set to NA, thus appearing as holes in the map. A: model with a weak prior for ϕ, B: model with an informative prior for ϕ. Averages were based on the 1,200 posterior distribution draws of each parameter.

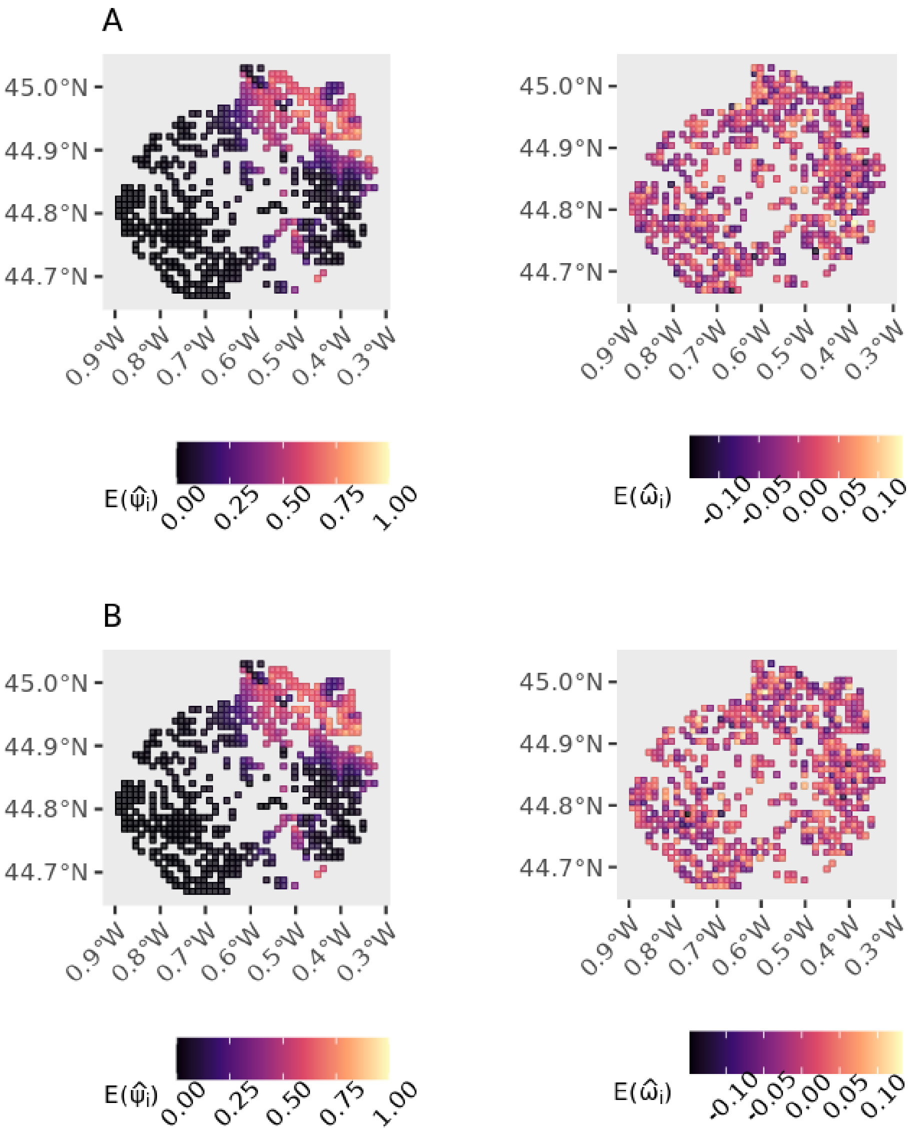

Figure C.2: Predicted occupancy $E(\hat{\psi}_i)$ of the large copper *L. dispar*, and values of the spatial random effect $E(\hat{\omega}_i)$ among the non-sampled cells in an area of 10 km around Bordeaux. Values for cells with butterfly records were set to NA, thus appearing as holes in the map. A: model with a weak prior for ϕ, B: model with an informative prior for ϕ. Averages were based on the 1,200 posterior distribution draws of each parameter.

D Supporting Information D - Sensitivity analyses using empirical data from the full Nouvelle-Aquitaine region

Table D.1: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to the common blue *P. icarus* occupancy data (full Nouvelle-Aquitaine data set). Results for models with 15 spatial neighbors, and weak or informative ϕ priors. PPO: Prior-Posterior Overlap. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, LAND= non-water land cover, GRASS: Cover of natural grasslands, OBS: Number of observers, MON: survey month. The mean, 95% Credible Intervals, and Rhat were calculated using the 1,200 posterior distribution draws of each parameter.

-	Weak prior, NNGP=15					Informative prior, NNGP=15				
Parameter	Mean	LCI	UCI	Rhat	Overlap	Mean	LCI	UCI	Rhat	Overlap
β_0	-0.414	-0.733	-0.111	1.100	18.1	-0.453	-0.865	-0.039	1.303	23.0
β_{LAT}	-0.094	-0.256	0.076	1.005	10.1	-0.090	-0.246	0.065	1.006	10.9
β_{LAT^2}	-0.247	-0.353	-0.146	1.007	7.3	-0.236	-0.344	-0.134	1.021	7.3
β_{LON}	0.053	-0.078	0.176	1.001	7.9	0.052	-0.068	0.175	1.026	8.9
β_{ELEV}	-0.306	-0.425	-0.183	1.064	7.9	-0.298	-0.416	-0.177	1.018	8.5
$\beta_{NON-WATER}$	0.374	-0.089	0.812	1.169	24.3	0.368	-0.107	0.832	1.029	26.9
$\beta_{NON-WATER^2}$	-0.321	-0.748	0.139	1.179	24.7	-0.321	-0.774	0.136	1.030	25.8
β_{GRASS}	-0.145	-0.628	0.374	1.007	26.6	-0.151	-0.651	0.353	1.008	28.6
β_{GRASS^2}	0.035	-0.051	0.117	1.004	5.4	0.035	-0.047	0.115	1.010	6.3
σ^2	4.695	3.826	5.671	1.159	4.7	4.453	3.454	5.579	1.129	7.7
ϕ	31.301	4.364	58.593	1.018	8.4	1.714	0.552	2.933	0.999	96.5
σ^2_T	0.362	0.175	0.764	1.031	50.9	0.366	0.178	0.692	1.092	53.6
ρ	0.048	-0.381	0.451	1.016	49.2	0.068	-0.359	0.475	1.062	47.4
α_0	-0.815	-0.917	-0.711	1.004	7.3	-0.807	-0.913	-0.706	1.001	7.4
α_{LAT}	0.119	0.053	0.183	1.021	5.1	0.118	0.048	0.186	1.002	4.9
α_{OBS}	0.034	0.029	0.039	1.022	0.8	0.034	0.029	0.040	1.013	0.9
α_{MON}	0.936	0.873	0.998	1.006	3.9	0.936	0.876	0.998	1.003	4.2
α_{MON^2}	-1.049	-1.127	-0.968	1.009	5.7	-1.052	-1.135	-0.971	1.001	5.8
$\alpha_{NON-WATER}$	0.048	0.012	0.085	1.002	3.1	0.048	0.011	0.085	1.041	3.2

Table D.2: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to the large copper *L. dispar* occupancy data (full Nouvelle-Aquitaine data set). Results for models with 15 spatial neighbors, and weak or informative ϕ priors. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, MAR=natural marshes, LAND= non-water land cover, OBS: Number of observers, MON: survey month. The mean, 95% Credible Intervals, and Rhat were calculated using the 1,200 posterior distribution draws of each parameter.

-	Weak prior, NNGP=15					Informative prior, NNGP=15				
Parameter	Mean	LCI	UCI	Rhat	Overlap	Mean	LCI	UCI	Rhat	Overlap
β_0	-8.043	-9.405	-7.198	3.903	0.0	-8.381	-9.259	-7.092	1.111	0.0
β_{LAT}	-0.414	-0.849	0.128	1.055	27.3	-0.369	-0.842	0.178	1.280	29.6
β_{LAT^2}	1.619	1.243	2.056	1.758	16.1	1.667	1.291	2.073	1.247	14.0
β_{LON}	1.998	1.574	2.480	1.985	14.7	2.082	1.614	2.514	1.413	12.7
β_{ELEV}	-5.488	-6.642	-4.437	2.211	0.9	-5.693	-6.617	-4.494	1.550	0.6
$\beta_{MARSHES}$	0.773	0.439	1.183	1.058	21.1	0.756	0.407	1.219	1.084	21.8
$\beta_{MARSHES^2}$	-0.100	-0.161	-0.051	1.056	4.2	-0.098	-0.160	-0.048	1.022	4.8
σ^2	14.297	10.729	19.946	3.075	0.7	14.331	9.397	19.585	1.470	0.8
ϕ	31.351	4.175	58.871	1.001	8.8	1.768	0.582	2.950	1.004	95.7
σ_T^2	1.415	0.581	3.304	1.544	52.6	1.255	0.532	2.760	1.113	53.0
ρ	0.253	-0.130	0.624	1.274	44.0	0.198	-0.186	0.526	1.070	41.4
α_0	-1.116	-1.470	-0.780	1.199	17.1	-1.145	-1.487	-0.809	1.130	17.4
α_{LAT}	-0.279	-0.475	-0.099	1.120	12.5	-0.299	-0.472	-0.143	1.272	11.2
α_{OBS}	0.053	0.038	0.066	1.041	1.4	0.053	0.039	0.067	1.008	1.4
α_{MON}	-0.150	-0.285	0.002	1.005	10.5	-0.143	-0.289	0.005	1.010	10.1
α_{MON^2}	-1.459	-1.674	-1.245	1.033	10.7	-1.462	-1.687	-1.249	1.026	10.6
$\alpha_{NON-WATER}$	0.185	0.099	0.269	1.054	6.2	0.180	0.087	0.275	1.053	7.1

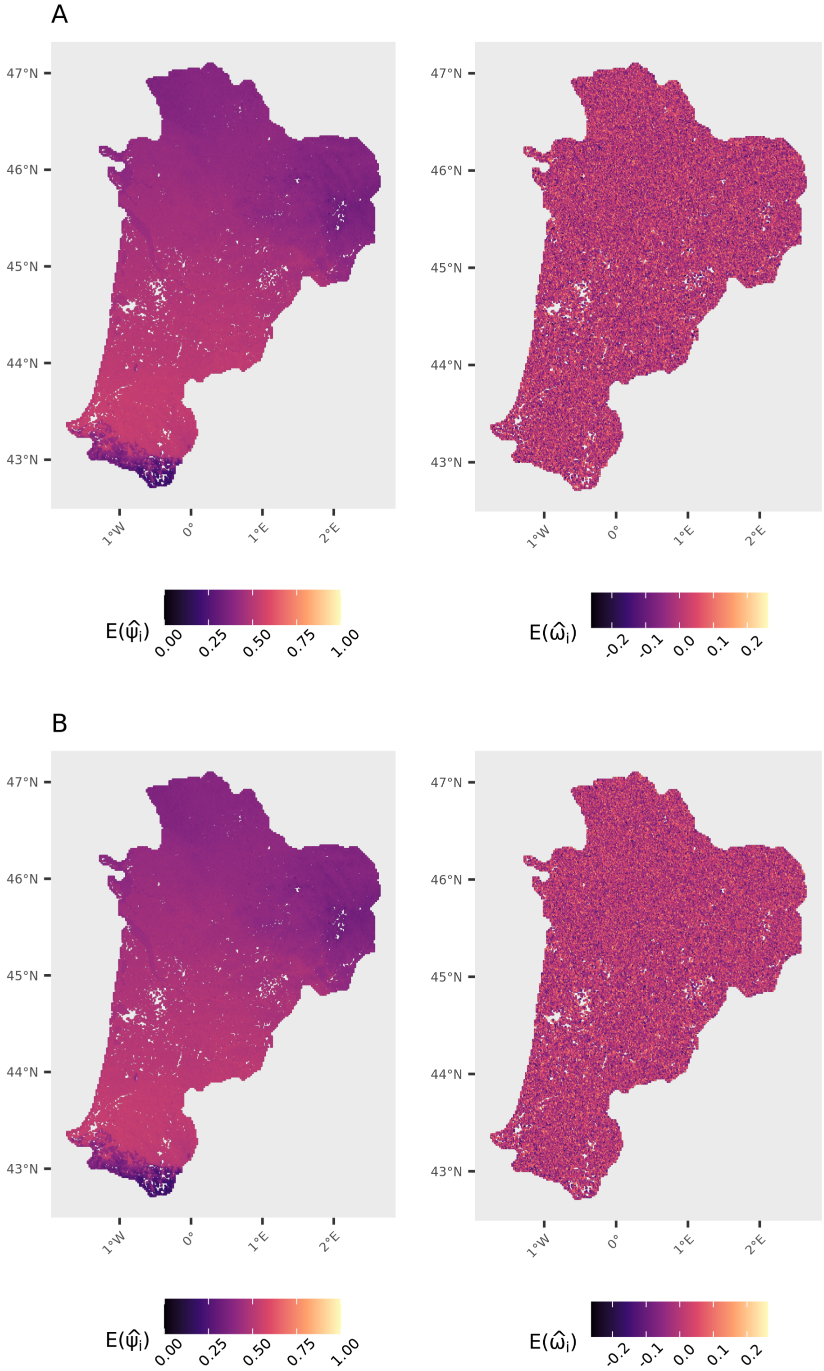

Figure D.1: Predicted occupancy $E(\hat{\psi}_i)$ of the common blue *P. icarus*, and values of the spatial random effect $\hat{\omega}_i$ among the non-sampled cells in Nouvelle-Aquitaine, Southwest France. Results from the model fitted to full occupancy data set. Values for cells with butterfly records were set to NA, thus appearing as holes in the map. A: model with a weak prior for ϕ, B: model with an informative prior for ϕ. Parameter values obtained from 1,200 posterior distribution draws.

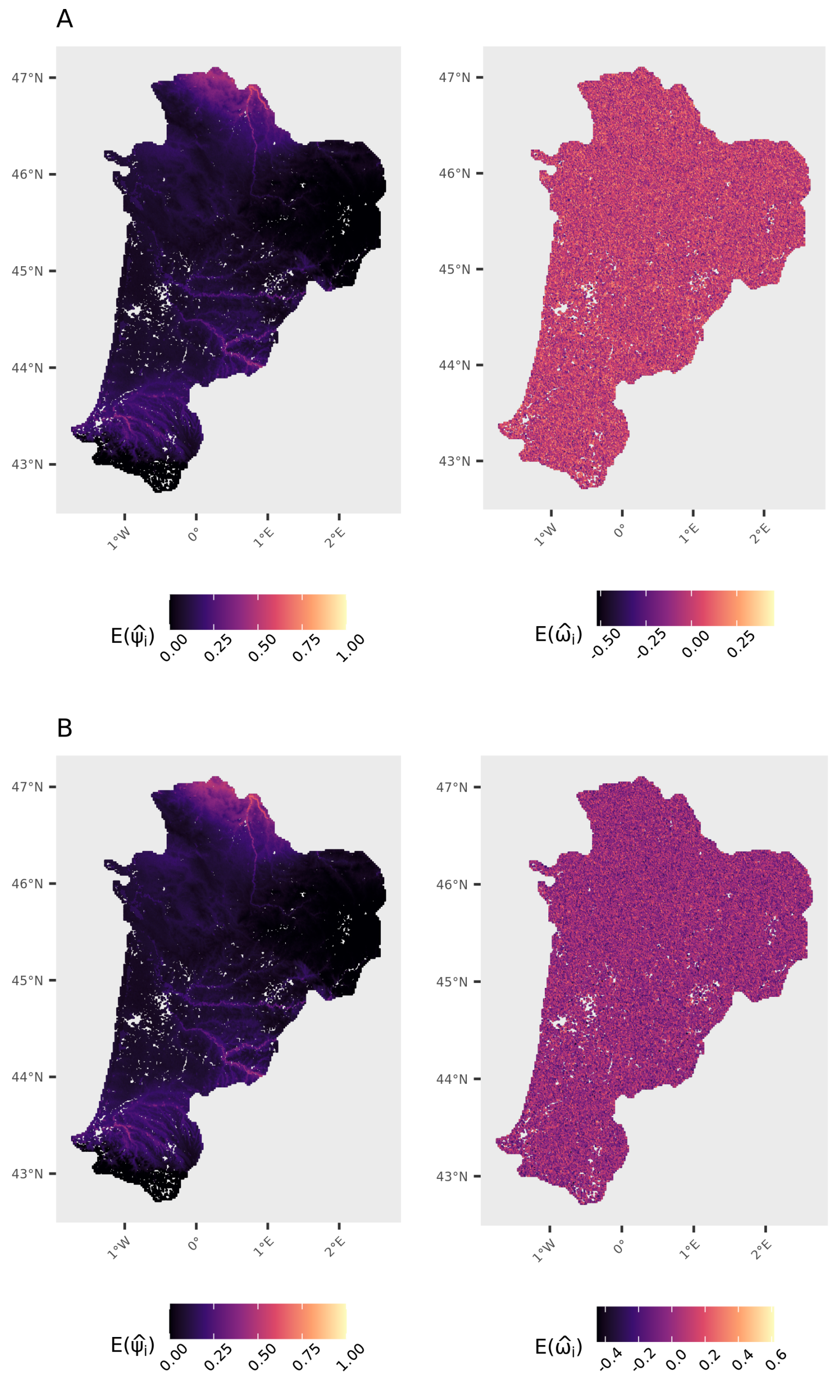

Figure D.2: Averages of predicted occupancy $\hat{\psi}_i$ of the large copper *L. dispar*, and values of the spatial random effect $E(\hat{\omega}_i)$ among the non-sampled cells in Nouvelle-Aquitaine, Southwest France. Results from the model fitted to full occupancy data set. Values for cells with butterfly records were set to NA, thus appearing as holes in the map. A: model with a weak prior for ϕ, B: model with an informative prior for ϕ. Parameter values obtained from 1,200 posterior distribution draws.

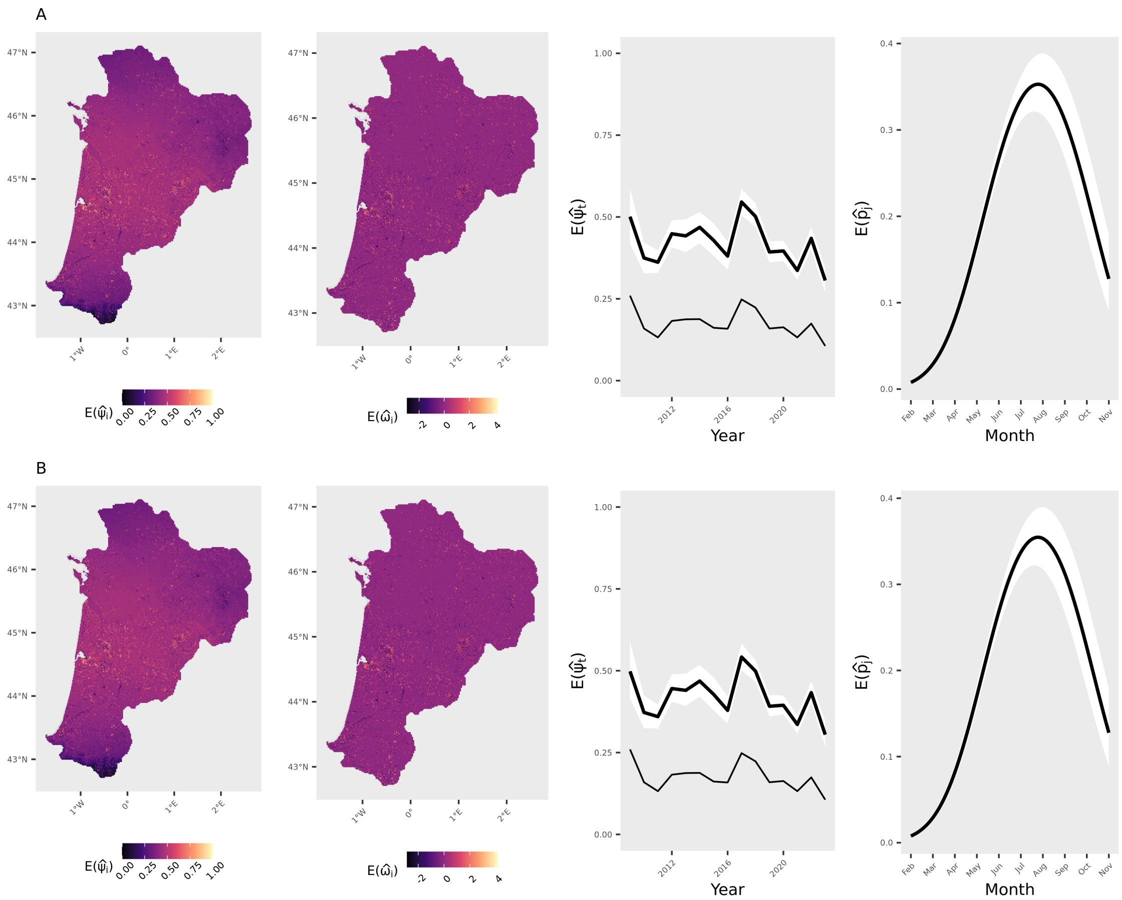

Figure D.3: Distribution of the common blue *Polyommatus icarus* in Nouvelle-Aquitaine, Southwest France, estimated by fitting the model to the full occupancy data set. (A) Results for the model with NNGP=15 and weakly informative prior for ϕ. (B) Results for the model with NNGP=15 and an informative prior for ϕ. Values plotted in the occupancy map (left) were calculated as $E(\hat{\psi}_i) = \frac{1}{T}\sum_{t=1}^{T} \ \hat{\psi}_{it}$, and refer to in-sample estimates for sites with data (cells with yellow borders) and out-of-sample predictions for missing cells (those with red borders). The spatial random effect $E(\hat{\omega}_i)$ was also obtained from in-sample estimates and out-of-sample predictions from the model. Trend plots show the proportion of occupied sites over the 15 years of data, calculated as $E(\hat{\psi}_t) = \frac{1}{I}\sum_{i=1}^{I} \ \hat{\psi}_{it}$. The naive occupancy trend (number of sites with detection divided by the total number of sampled sites per year) is depicted by the thinner line. Detection probability over months (secondary periods) was also obtained from in-sample predictions. Parameter values obtained from 1,200 posterior distribution draws.

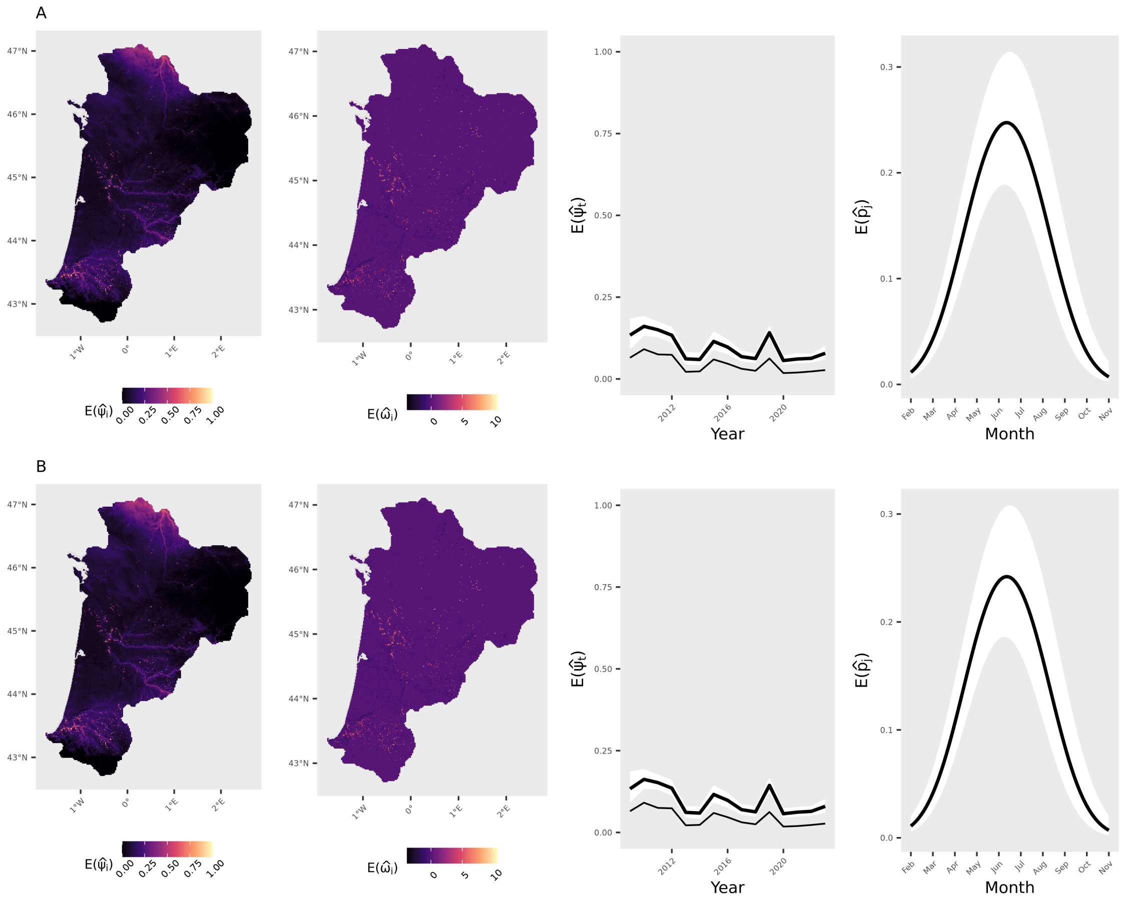

Figure D.4: Distribution of the large copper *Lycaena dispar* in Nouvelle-Aquitaine, Southwest France, estimated by fitting the model to the full occupancy data set. (A) Results for the model with NNGP=15 and weakly informative prior for ϕ. (B) Results for the model with NNGP=15 and an informative prior for ϕ. Values plotted in the occupancy map (left) were calculated as $E(\hat{\psi}_i) = \frac{1}{T}\sum_{t=1}^{T} \hat{\psi}_{it}$, and refer to in-sample estimates for sites with data (cells with yellow borders) and out-of-sample predictions for missing cells (those with red borders). The spatial random effect $E(\hat{\omega}_i)$ was also obtained from in-sample estimates and out-of-sample predictions from the model. Trend plots show the proportion of occupied sites over the 15 years of data, calculated as $E(\hat{\psi}_t) = \frac{1}{I}\sum_{i=1}^{I} \hat{\psi}_{it}$. The naive occupancy trend (number of sites with detection divided by the total number of sampled sites per year) is depicted by the thinner line. Detection probability over months (secondary periods) was also obtained from in-sample predictions. Parameter values obtained from 1,200 posterior distribution draws.

References

Doser, J.W. & Stoudt, S. (2024). "Fractional replication" in single-visit multi-season occupancy models: Impacts of spatiotemporal autocorrelation on identifiability. *Methods in Ecology and Evolution*, 15, 358–372.

Gourvil, P.Y. & Sannier, M. (2020). *Atlas des papillons de jour d'Aquitaine.* Inventaires & biodiversité. Biotope, Mèze.